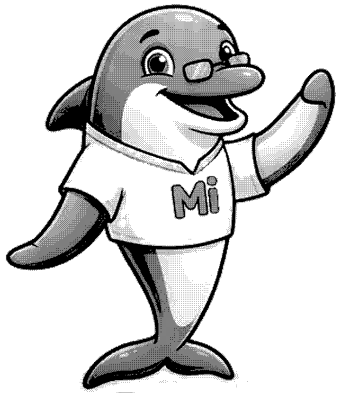
Math Inquirers

# Extension Workbook

## with answers

## Level 2

Self-study | Ages 7-9




My name _____

# About this book

**Math Inquirers Extension Workbook, Level 2**
Self-study practice with answers.


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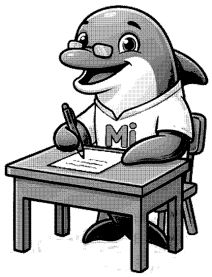
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- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

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# Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

**How to use this book**

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

**For the grown-up**

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 79: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Beyond Level 2 is ahead of school. You may skip it.

**The four page names**

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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# My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

| Week | Unit | Session 1<br>Learn it, Start here | Session 2<br>Start here | Session 3<br>Think harder, Check yourself | Weekend page<br>Go further | Done |
|---|---|---|---|---|---|---|
| 1 | Digit detectives | 6-7 | 8-9 | 10 and 12 | 11 | ☐ |
| 2 | Missing digit mysteries | 13-14 | 15-16 | 17 and 19 | 18 | ☐ |
| 3 | Legs, wheels and arrays | 20-21 | 22-23 | 24 and 26 | 25 | ☐ |
| 4 | Pattern machines | 27-28 | 29-30 | 31 and 33 | 32 | ☐ |
| 5 | Mystery number machines | 34-35 | 36-37 | 38 and 40 | 39 | ☐ |
| 6 | Market day | 41-42 | 43-44 | 45 and 47 | 46 | ☐ |
| 7 | Fair shares | 48-49 | 50-51 | 52 and 54 | 53 | ☐ |
| 8 | Shape builders | 55-56 | 57-58 | 59 and 61 | 60 | ☐ |
| 9 | Measure masters | 62-63 | 64-65 | 66 and 68 | 67 | ☐ |
| 10 | Puzzle day | 69-70 | 71-72 | 73 and 75 | 74 | ☐ |
| 11 | Catch-up week | Repeat any Check yourself where your score was 2 or less. | | | | ☐ |

**Tips for the grown-up**

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

# Learn it: Numbers to 1000

**My goal. I can read and build 3-digit numbers. I can list every number from a set of digit cards and use patterns on a 1000 chart.**

A 3-digit number has hundreds, tens and ones. Where a digit sits tells you what it is worth.

## Worked example

**Use the digit cards 4, 1 and 7 once each. List all the 3-digit numbers the cards can make. Which is the greatest? Which is the smallest?**




1. **Put 1 in the hundreds place. Swap the other cards: 147 and 174.**
2. **Put 4 in the hundreds place: 417 and 471.**
3. **Put 7 in the hundreds place: 714 and 741.**
4. **Check: 3 cards start the list, and each starts 2 numbers. 3 x 2 = 6 numbers.**

**Answer: 147, 174, 417, 471, 714, 741. Greatest 741, smallest 147.**

## How to list in order

1. Fix one card in the hundreds place.
2. Swap the other cards.
3. Do the same for each card.
4. Count: cards x 2 = how many numbers?

## Place value

- 624 is 6 hundreds, 2 tens and 4 ones.
- 10 tens make 1 hundred, so 12 tens is 120.
- 0 cannot be the hundreds digit. 047 is not a 3-digit number.

## Your turn

Answer at the back of the book.

Try **Use the cards 9, 0 and 3 once each. How many 3-digit numbers can you make?** ☐

Name _____

## Start here

Do these first. Take your time.

**1** **Place value. Fill in the blanks.**

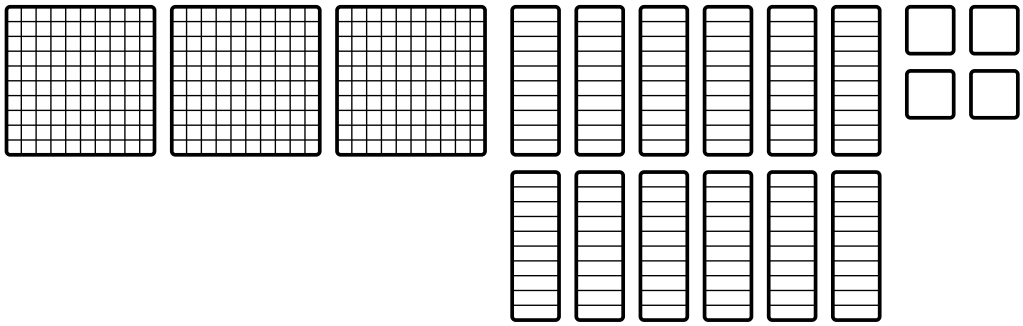
(a) 3 hundreds, 5 tens and 6 ones = ☐

(b) 8 hundreds, 0 tens and 4 ones = ☐

(c) 590 = 5 hundreds and ☐ tens

(d) In 837, the digit 3 is worth ☐

**2** **What number do the place-value blocks show?** ☐



■ = 100 ▮ = 10 □ = 1

**3** **Circle the correct sign: <, = or >.**

(a) 745 < = > 754 (b) 306 < = > 296

(c) 5 hundreds 2 tens < = > 502

(d) 88 tens < = > 880

**4** **Arrange the numbers in order. Begin with the smallest.**

**508, 85, 580, 805**

_____

## Start here

Keep going. Check your work.

**5** **Use the cards 8, 1 and 5 once each. Fill the table with all the 3-digit numbers the cards can make.**




| Start with 1 | Start with 5 | Start with 8 |
|---|---|---|
| | | |
| | | |

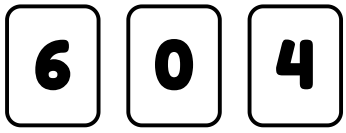
**6** **Use your list from the question before.**

**(a)** From your list: greatest ☐, smallest ☐.

**(b)** Which number is closest to 600? ☐

**(c)** How many of your numbers are odd? ☐

**7** **Use the cards 6, 0 and 4 once each.**




**(a)** List all the 3-digit numbers the cards can make.

_____

**(b)** How many numbers did you write? ☐

**8** **Who am I? Write each number.**

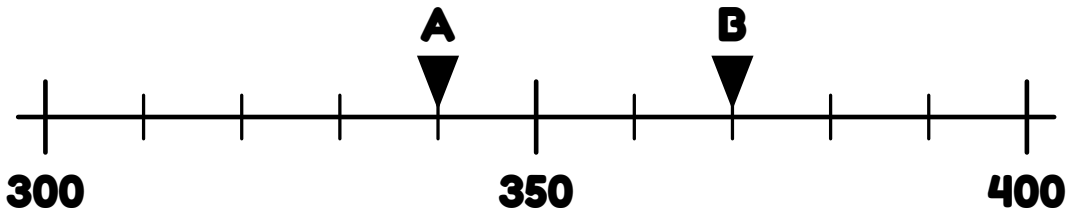
**(a)** My number has 3 digits. The hundreds digit is 6. The ones digit is 2 less than the tens digit. The digits add up to 14. ☐

**(b)** My number is odd and between 300 and 310. Its digits add up to 8. ☐

## Start here

Nearly there. You can do it.

**9** **Look at the number line. It goes from 300 to 400.**




(a) Arrow A shows ☐ (b) Arrow B shows ☐

(c) B is ☐ more than A.

**10** **Each piece below was cut out of a 1000 chart. A row is 10 numbers long. Write the missing numbers.**

**Piece A**

| | | |
|---|---|---|
| 571 | 572 | |
| | 582 | 583 |
| 591 | | |

**Piece B**

| | | |
|---|---|---|
| | | |
| | 645 | |
| | | |

(a) Piece A: write the 4 missing numbers in the empty squares.

(b) Piece B: write the 4 missing numbers in the empty squares.

(c) On the chart, the square 3 below 612 holds ☐

**11** **Circle True or False.**

(a) 5 tens and 12 ones is the same as 62. True False

(b) 7 hundreds is the same as 700 tens. True False

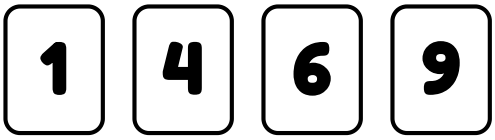
(c) In 909, the two 9s are worth the same. True False

(d) 10 hundreds is the same as 1000. True False

## Think harder

Hints are upside down. Try first!

**1** **Use the cards 1, 4, 6 and 9. Pick 3 different cards each time to make 3-digit numbers.**




**(a)** How many even numbers can you make? ☐

**(b)** How many of those even numbers are greater than 600? ☐

Working: _____

**Hint.** Fix the ones digit first. An even number ends in 4 or 6.

**2** **Find my number. It is odd and has 3 digits. My tens digit is 0. My digits add up to 11.**

**(a)** The numbers I could be:

_____

**(b)** Which of them is closest to 500? ☐

**Hint.** The hundreds and ones digits add up to 11. The ones digit is odd.

**3** **Ali writes the numbers 1 to 100 in a row. Then he writes 101 to 200 in a row.**

**(a)** How often does Ali write the digit 7 from 1 to 100? ☐

**(b)** How often does Ali write the digit 7 from 1 to 200? ☐

Working: _____

**Hint.** Count the 7s in the ones place. Then count the 7s in the tens place.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: Thousands

**1** **10 hundreds make 1 thousand. We write 1 thousand as 1000.**

(a) 4 thousands, 2 hundreds, 0 tens and 7 ones = ☐

(b) 1000 more than 2650 = ☐

(c) 25 hundreds = ☐

**Hint.** A thousand is 10 hundreds. Write the thousands digit first.

**2** **A mirror number reads the same forwards and backwards, such as 262 and 909. How many 3-digit mirror numbers are there?** ☐

Working: _____

**Hint.** The ones digit must match the hundreds digit. Count the choices.

**3** **My number has 3 digits and my tens digit is 5. My digits add up to 12. If my hundreds digit and ones digit swap places, the new number is 297 more than me. What is my number?** ☐

Working: _____

**Hint.** The hundreds and ones digits add up to 7. Try pairs and swap them.

**4** **Make your own riddle about the number 372. Write 3 clues so that only 372 fits. Many answers are right.**

Clue 1: _____

Clue 2: _____

Clue 3: _____

**Hint.** Say how many numbers still fit after each clue.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **7 hundreds and 13 tens =** ☐

Working: _____

**2** **Use the cards 5, 0 and 8 once each. Write the greatest even 3-digit number.** ☐

Working: _____

**3** **Count back in 10s. 312, 302, 292,** ☐ , ☐

Working: _____

**4** **Count the odd numbers between 530 and 542.** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

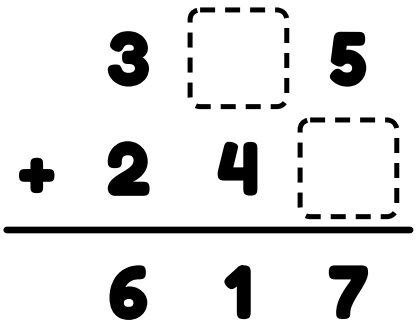
# Learn it: Adding and subtracting to 1000

**My goal.** I can find missing digits in 3-digit sums. I can balance number sentences and solve magic squares and triangles.

A missing digit is a mystery to solve. Start at the ones column and ask: did I carry? The = sign means "is the same as".

## Worked example

**Find the missing digits in this sum.**




1. **Ones: 5 + ? must end in 7. So the missing digit is 2.**
2. **Tens: ? + 4 must end in 1. 11 is possible: 7 + 4 = 11. Write 1 and carry 1.**
3. **Hundreds: 3 + 2 + 1 carried = 6. It matches.**
4. **Check: 375 + 242 = 617.**

**Answer: The missing digits are 7 and 2.**

## How to find a missing digit

1. Start at the ones column.
2. Ask: did I carry?
3. Move left, one column at a time.
4. Check by adding the whole numbers.

## Balance

- = means "is the same as".
- 260 + 150 = 410, so 300 + [ ] must be 410.
- The box is 110.

## Your turn

Answer at the back of the book.

Try **Find the missing digit. 5?4 + 337 = 891 The digit is** ☐

## Start here

Do these first. Take your time.

**1** **Find the missing digits. Each sum has 2. Write them in the answer boxes, top number first.**

| (a) | (b) | (c) |
|---|---|---|
| 2 □ 4 | 4 □ 8 | □ 6 5 |
| + 3 5 □ | + □ 2 6 | + 2 □ 8 |
| 6 0 3 | 8 7 4 | 7 0 3 |

(a) Sum (a): ☐☐ (b) Sum (b): ☐☐

(c) Sum (c): ☐☐

**2** **Find the missing digits in these subtractions. Check each one by adding the answer to the number taken away. Write the digits top number first.**

| (a) | (b) |
|---|---|
| 8 □ 5 | 5 □ 2 |
| - 2 6 □ | - □ 4 7 |
| 5 6 8 | 2 6 5 |

(a) Sum (a): ☐☐ (b) Sum (b): ☐☐

**3** **Both sides must have the same value. Fill in the boxes.**

(a) 270 + 140 = 300 + ☐ (b) 840 - 300 = ☐ - 200

(c) 360 + ☐ = 400 + 150

(d) 199 + 250 = 200 + 249 True False

## Start here

Keep going. Check your work.

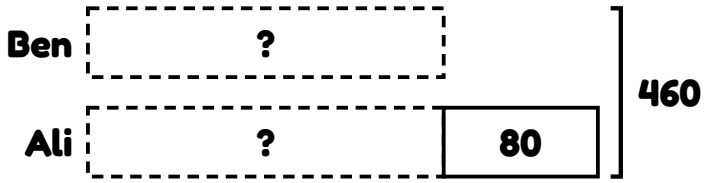
**4** **This is a magic square. Every row, column and corner-to-corner line has the same total.**

| | | |
|---|---|---|
| 100 | | 300 |
| | 250 | |
| 200 | | 400 |

(a) The magic total is ☐

(b) Fill in the 4 empty squares.

**5** **Ali and Ben have 460 stickers altogether. Ali has 80 more than Ben.**




(a) Take away the extra 80: 460 − 80 = ☐

(b) Ben has ☐ stickers.

(c) Ali has ☐ stickers.

**6** **Ben wrote 614 − 278 = 346. He checks by adding.**

(a) 346 + 278 = ☐

(b) Is Ben right? Yes No

(c) The correct answer is ☐

## Start here

Nearly there. You can do it.

**7** **Ali had 345 stickers. He gave some away. Now he has 128 stickers.**

**(1) 345 + [ ] = 128** **(2) 345 - [ ] = 128**

**(3) [ ] - 345 = 128** **(4) 128 - [ ] = 345**

(a) Which number sentence matches the story?

(1) (2) (3) (4)

(b) How many stickers did Ali give away? ☐

**8** **Mei had 500 beads. She used some for a necklace and 135 for a bracelet. 210 beads were left. How many beads did she use for the necklace?** ☐

Working: _____

**9** **Add or subtract.**

(a) 368 + 457 = ☐ (b) 702 - 385 = ☐

(c) 249 + 394 = ☐ (d) 910 - 466 = ☐

**10** **Make 1000. Fill in the missing number.**

(a) 340 + ☐ = 1000 (b) 485 + ☐ = 1000

(c) 1000 - 755 = ☐ (d) ☐ + 75 = 1000

**11** **Lily's book has 500 pages. She has read 287 pages.**

(a) How many pages are left? ☐

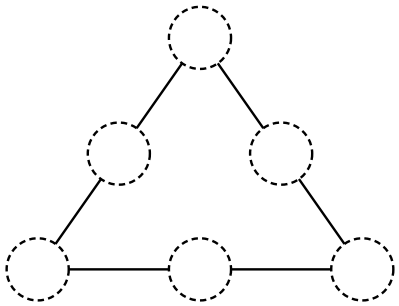
(b) Then she reads 120 more pages. How many pages are left now? ☐

## Think harder

Hints are upside down. Try first!

### 1

**Fill the 6 circles with 120, 135, 150, 165, 180 and 195. Use each number once. Each side must add to 450.**



(a) Corners: ☐, ☐, ☐.

(b) Sides add up to 495. Corners: ☐, ☐, ☐.

**Hint.** Which numbers are used in 2 sides?

### 2

**Circle True or False. Do not add both sides.**

(a) 468 + 199 = 467 + 200 True False

(b) 350 + 240 = 340 + 260 True False

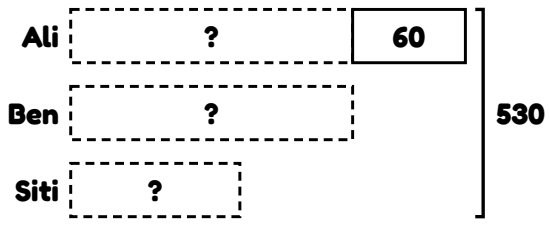
(c) 502 + 99 = 500 + 101 True False

(d) 275 + 275 = 250 + 310 True False

**Hint.** Move the same amount from one number to the other.

### 3

**Ali, Ben and Siti have 530 stickers altogether. Ali has 60 more than Ben. Siti has 40 fewer than Ben.**




(a) Ben has ☐ stickers. (b) Ali has ☐ stickers.

(c) Siti has ☐ stickers.

**Hint.** Make all 3 bars the same length as Ben's bar.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Ali and Mei have 540 stickers together. Mei and Ravi have 610 together. Ali and Ravi have 450 together. How many stickers does each child have?**

(a) Ali has ☐ (b) Mei has ☐ (c) Ravi has ☐

**Hint.** Add the 3 totals, then halve. That is all 3 children together.

★ Beyond Level 2: Patterns in subtraction

**2** **Reverse a 3-digit number by swapping its first digit and last digit. Then take the smaller number from the greater.**

(a) 742 - 247 = ☐ 831 - 138 = ☐

(b) What is the middle digit of both answers? ☐

(c) Use the pattern: 954 - 459 = ☐

**Hint.** Look at the middle digit. Add the digits of each answer.

**3** **Make your own missing-digit sum. Add two 3-digit numbers and hide 2 digits. Only ONE pair of digits may work. Many answers are right.**

My sum: _____

**Hint.** Start with a full sum. Then hide a digit in each number.

**4** **Mei adds 198 to her number. Ali takes 102 away from 900. They get the same answer.**

(a) Mei's number is ☐

(b) Ravi adds 299 to his number. Siti takes 201 away from 1000. Same answer. Ravi's number is ☐

**Hint.** Find the answer Ali gets first.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Write the 2 missing digits.**

$$\begin{array}{r} 2\,\square\,6 \\ +\ 4\ 3\ \square \\ \hline 6\ 8\ 9 \end{array}$$

**2** **530 + 160 = 600 + ☐**

Working: _____

**3** **Ravi had 800 stickers. He gave 175 to Mei and some to Ben. 350 were left. How many did Ben get?** ☐

Working: _____

**4** **Two numbers add up to 420. One is 60 more. What is the smaller number?** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# Learn it: Multiplication and division

**My goal. I can read an array and write its sentences. I can find how many of each from a table. I can share with leftovers.**

Equal groups are everywhere: legs, wheels, rows of counters. An array shows rows of equal groups. When you share, some may be left over.

## Worked example

**A yard has 6 animals, ducks (2 legs) and dogs (4 legs). There are 16 legs altogether. How many ducks and how many dogs?**

| Try | 1 | 2 | 3 |
|---|---|---|---|
| Ducks | 6 | 5 | 4 |
| Dogs | 0 | 1 | 2 |
| Legs | 12 | 14 | 16 |

1. Start with all ducks: 6 ducks have 12 legs.
2. Swap 1 duck for 1 dog. That adds 2 legs: 14.
3. Swap again: 16 legs. Stop!
4. Check: 4 ducks and 2 dogs. 8 + 8 = 16 legs and 4 + 2 = 6 animals.

**Answer: 4 ducks and 2 dogs.**

## How to find how many of each

1. Start with all of one kind.
2. Swap one at a time. Write the total.
3. Stop at the number you need.
4. Check both clues.

## Arrays and sharing

- 3 rows of 4: 3 × 4 = 12 and 4 × 3 = 12.
- 12 ÷ 3 = 4 and 12 ÷ 4 = 3.
- 17 shared among 5: 3 each and 2 left over.

## Your turn

Answer at the back of the book.

Try **There are 3 animals, ducks and dogs. They have 10 legs altogether.**

**How many dogs are there?** ☐

## Start here

Do these first. Take your time.

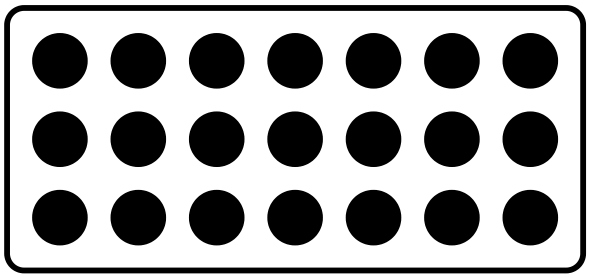
**1** **Look at the array of counters.**

(a) ☐ × ☐ = ☐

☐ × ☐ = ☐

(b) ☐ ÷ ☐ = ☐

☐ ÷ ☐ = ☐

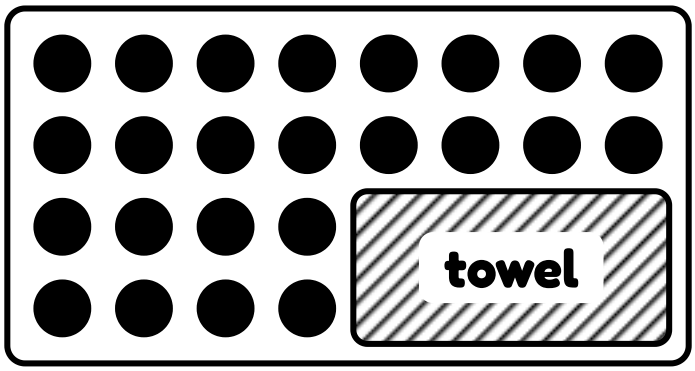


**2** **A baking tray has the same number of cookies in every row. A towel covers some of the cookies.**

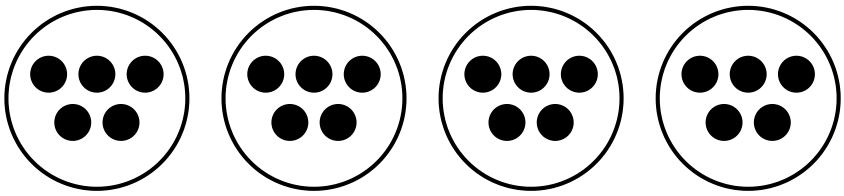
(a) How many cookies are in each row? ☐

(b) How many cookies are on the whole tray? ☐

(c) How many are under the towel? ☐




**3** **Look at the plates. Circle True or False.**



(a) There are 4 groups of 5. True False

(b) 5 + 5 + 5 = 15 shows all the counters. True False

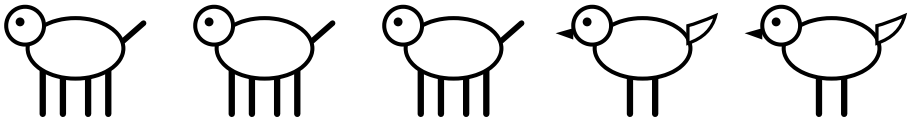
(c) 4 × 5 = 20 matches the picture. True False

(d) There are 5 plates with 4 counters on each. True False

## Start here

Keep going. Check your work.

**4** **Look at the animals.**



dog dog dog duck duck

(a) 3 dogs have ☐ legs.

(b) 2 ducks have ☐ legs.

(c) All the animals have ☐ legs.

**5** **A pen has 8 animals, ducks and rabbits. A duck has 2 legs and a rabbit has 4 legs. There are 26 legs altogether. Start with all ducks. Swap 1 duck for 1 rabbit each time.**

| Ducks | 8 | 7 | 6 | 5 | 4 | 3 |
|---|---|---|---|---|---|---|
| Rabbits | 0 | 1 | 2 | 3 | 4 | 5 |
| Legs | 16 | 18 | | | | |

(a) Write the 4 missing numbers of legs in the table.

(b) There are ☐ ducks and ☐ rabbits.

**6** **Use groups to find the answer.**

(a) 6 boxes have 4 pencils in each box. How many pencils? ☐

(b) 30 sweets go equally into 5 bags. How many in each bag? ☐

(c) 45 stickers go in packs of 5. How many packs? ☐

(d) How many wheels do 6 cars have? Each car has 4 wheels. ☐

## Start here

Nearly there. You can do it.

**7** **Siti has 37 biscuits. She packs 5 biscuits in each box.**

(a) She fills as many boxes as she can. How many boxes? ☐

(b) How many biscuits are left over? ☐

(c) How many more biscuits does she need to fill 1 more box? ☐

**8** **47 people go to a farm. Each minibus holds 10 people.**

(a) How many minibuses must they use? ☐

(b) How many seats will be empty? ☐

**9** **Mr Lim puts 18 plant pots in equal rows. There must be 2 rows or more and 2 pots or more in each row. Write the pots in each row. Write no if it does not work.**

| Rows | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|
| Pots | | | | | | | | |

(a) Fill in the table.

(b) How many different ways are there? ☐

**10** **Share 24 sweets equally.**

(a) Among 3 children, each child gets ☐

(b) Among 4 children, each child gets ☐

(c) Among 10 children, each child gets ☐. Left over: ☐.

(d) Siti shares them with her 3 friends. Each person gets ☐

## Think harder

Hints are upside down. Try first!

**1** **A yard has scooters (2 wheels) and trikes (3 wheels), at least 1 of each. Altogether the vehicles have 21 wheels. List ALL the ways.**

| Trikes | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Their wheels | 3 | 6 | 9 | 12 | 15 | 18 |
| Wheels left | | | | | | |

(a) How many different ways are there? ☐

(b) What is the fewest number of vehicles you can have? ☐

**Hint.** Scooters use 2 wheels each. So wheels left must be even.

**2** **A pen has 6 animals. Each is a hen (2 legs) or a pig (4 legs). All 6 may be hens. Mei counts the legs. Could she count this number?**

(a) 18 legs Yes No (b) 25 legs Yes No

(c) 26 legs Yes No (d) 12 legs Yes No

Why 25 is not possible: _____

**Hint.** All hens is 12 legs. Each swap adds 2 legs. All pigs is 24.

**3** **I am a number between 30 and 50. Put me in groups of 3 and nothing is left. Put me in groups of 4 and 1 is left. Put me in groups of 5 and 3 are left. What is my number?** ☐

Working: _____

**Hint.** List the numbers that go into groups of 3 first.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: The 6 times table

**1** **The 6 times table. 6 × 5 = 30, and 6 more is 6 × 6.**

**(a)** 6 × 6 = ☐ **(b)** 6 × 8 = ☐ **(c)** 54 ÷ 6 = ☐

**(d)** 6 × 7 = ☐

**Hint.** Count on in 6s from 30. Or double the 3 times table.

**2** **Mei has 9 bags. Some bags hold 3 apples. The other bags hold 5 apples. She has 35 apples altogether.**

**(a)** There are ☐ bags of 3 apples and ☐ bags of 5 apples.

**(b)** Could 9 bags hold 40 apples? Yes No

Why (b): _____

**Hint.** Start with all bags of 3. Each swap adds 2 apples.

**3** **Make your own puzzle. Use stools (3 legs) and chairs (4 legs). Choose how many pieces and how many legs, so that there is only ONE answer. Many answers are right.**

My puzzle: _____

My answer: _____

**Hint.** Pick the answer first. Then count the legs.

**4** **Ali has 24 counters. He makes arrays with 2 rows or more and 2 counters or more in each row. A 2 by 12 array and a 12 by 2 array are different. How many different arrays can he make?** ☐

Working: _____

**Hint.** Try 2 rows, 3 rows, 4 rows … up to 12 rows.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **A tray has 3 rows with 8 cups in each row. How many cups are there?**

☐

Working: _____

**2** **How many legs do 5 dogs and 4 ducks have altogether?** ☐

Working: _____

**3** **47 pencils go in boxes of 10. Full boxes:** ☐ **Left over:** ☐

Working: _____

**4** **9 animals, birds (2 legs) and rabbits (4 legs), have 24 legs. How many rabbits?** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

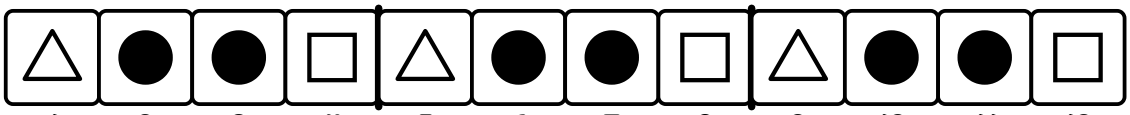
# Learn it: Patterns with multiplication

**My goal.** I can find far places in a repeating pattern. I can use a table and a rule for a growing pattern. I can find numbers that two counting patterns share.

A repeating pattern is one group again and again. The last shape of each group is in a times table. A growing pattern goes up by the same number each step. A table helps.

## Worked example

**This pattern repeats again and again. What is the 26th shape?**




1. One group has 4 shapes: triangle, circle, circle, square.
2. Groups end in the 4 times table: 4, 8, 12, 16, 20, 24.
3. Place 24 is the last shape of a group, a square.
4. Place 25 starts again with a triangle. Place 26 is a circle.

**Answer: The 26th shape is a circle.**

## How to find a far place

1. Find the size of one group.
2. Count in groups to near your place.
3. That place is the LAST shape of a group.
4. Count on to your place.

## Growing patterns

- Each step adds the same number.
- A table shows the steps and the rule.
- Check: is the number in the pattern?

## Your turn

Answer at the back of the book.

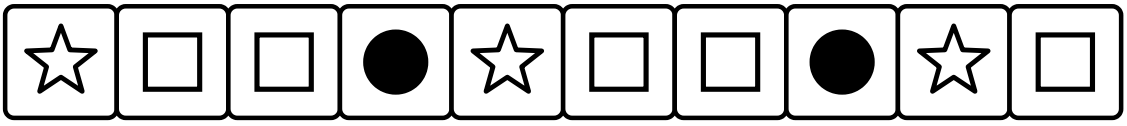
**Try** **The pattern star, square, square repeats again and again. What is the 14th shape?** star square

## Start here

Do these first. Take your time.

**1** Each step takes away 5. Write the next numbers. 60, 55, 50, ☐, ☐, ☐

**2** This pattern repeats. Here are places 1 to 10.




(a) Place 16 is a star square circle

(b) Place 22 is a star square circle

(c) Place 33 is a star square circle

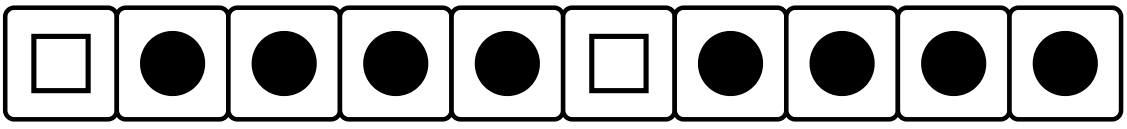
**3** Ali writes the letters of CHESS again and again.

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| C | H | E | S | S | C | H | E | S | S |

(a) The 17th letter is ☐ (b) The 6th E is in place ☐

(c) In 30 letters there are ☐ letters S.

**4** A bead string has 1 square bead, then 4 round beads, again and again. Here are the first 10 beads.



(a) The string has 8 square beads. How many round beads are there? ☐

(b) How many beads are on the string altogether? ☐

## Start here

Keep going. Check your work.

**5** **Each pizza is cut into 4 slices.**

| Pizzas | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Slices | 4 | | | | | |

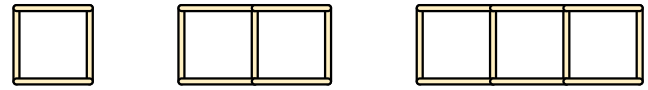
**(a)** Fill in the table.

**(b)** 10 pizzas have ☐ slices.

**(c)** 28 slices make ☐ pizzas.

**(d)** 14 slices make ☐ pizzas and ☐ slices are left over.

**6** **Sticks make a chain of squares. 1 square uses 4 sticks, 2 squares use 7 sticks and 3 squares use 10 sticks.**



1 square 2 squares 3 squares

**(a)** 4 squares use ☐ sticks.

**(b)** 6 squares use ☐ sticks.

**(c)** 8 squares use ☐ sticks.

**7** **Siti counts in 3s from 3. Ravi counts in 5s from 5. Write each list up to 40 on the lines below the questions.**

**(a)** The numbers they both say are:

_____

**(b)** The next number they both say is ☐

Siti: 3, 6, 9, _____

Ravi: 5, 10, 15, _____

## Start here

Nearly there. You can do it.

**8** **Find the rule. Then write the next 2 numbers.**

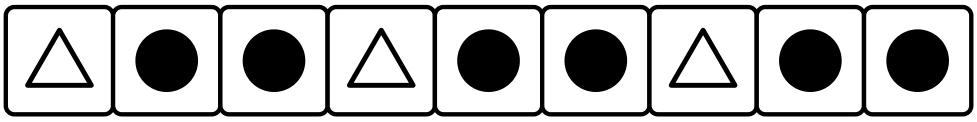
(a) 7, 12, 17, 22, ☐, ☐ The rule is

add 4 add 5 add 7

(b) 50, 44, 38, ☐, ☐ The rule is

subtract 5 subtract 6 subtract 7

**9** **A pattern goes triangle, circle, circle, again and again. Here are places 1 to 9.**



1 2 3 4 5 6 7 8 9

(a) 24 shapes make ☐ groups of 3.

(b) There are ☐ circles in the first 24 shapes.

(c) There are ☐ triangles in the first 24 shapes.

**10** **Towers of cubes grow in the same way. Tower 1 is 5 cubes tall. Tower 2 is 7 cubes tall. Tower 3 is 9 cubes tall.**

| Tower | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Cubes | 5 | 7 | 9 | | | |

(a) Fill in the table.

(b) Tower 10 has ☐ cubes.

(c) Tower ☐ has 19 cubes.

Working: _____

## Think harder

Hints are upside down. Try first!

**1** **A beetle starts at 0. It hops 3, then 1, then 3, then 1 … It lands on 3, 4, 7, 8, 11 …**

(a) Will it land on 27? Yes No

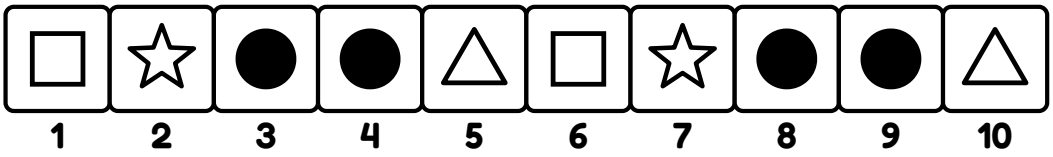
(b) Will it land on 26? Yes No

(c) Will it land on 40? Yes No

Working: _____

**Hint.** Write the landing places. Look at the 4 times table.

**2** **This pattern repeats. Here are places 1 to 10. Could a star be in this place?**




(a) Place 17 Yes No (b) Place 22 Yes No

(c) Place 35 Yes No (d) Place 47 Yes No

**Hint.** Stars are 2 more than a number in the 5 times table.

**3** **A toy train has an engine and carriages. The engine holds 2 children. Each carriage holds 5 children.**

| Carriages | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Children | 2 | 7 | 12 | |

(a) 4 carriages hold ☐ children.

(b) 8 carriages hold ☐ children.

(c) 37 children fill ☐ carriages.

**Hint.** Each carriage adds 5. The engine is extra.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: Square numbers

**1** **1, 4, 9, 16 are square numbers: 1 × 1, 2 × 2, 3 × 3, 4 × 4.**

(a) 6 × 6 = ☐

(b) The next 2 square numbers after 25 are ☐ and ☐.

(c) The 10th square number is ☐

**Hint.** A square number is a number times itself.

**2** **Frog A lands on 4, 8, 12 … (jumps of 4). Frog B starts on 1 and lands on 4, 7, 10 … (jumps of 3).**

(a) Up to 30, they both land on ☐, ☐ and ☐.

(b) The next number they both land on is ☐

(c) Frog C starts at 2 and jumps 4. Do A and C ever land on the same number? Yes No

**Hint.** They land together every 12. Why 12?

**3** **The pattern star, circle, circle, square, square repeats again and again. Ravi says place 40 is a star. Siti says place 40 is a square.**

(a) Who is right? Ravi Siti

(b) Place 43 is a star circle square

**Hint.** Place 40 is in the 5 times table. It is the last shape.

**4** **Make your own pattern of 4 shapes. It repeats again and again. Many answers are right.**

Draw the first 8 shapes:

Place 30 is a: _____

**Hint.** Place 28 is the last shape of a group. Count on from there.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **The pattern star, circle, square repeats again and again. The 20th shape is a star circle square**

Working: _____

**2** **Each box has 5 pens. 9 boxes have ☐ pens.**

Working: _____

**3** **Count in 4s from 4 and in 5s from 5, up to 40. How many numbers are in both lists? ☐**

Working: _____

**4** **Tower 1 has 3 cubes, tower 2 has 7 and tower 3 has 11. Tower 6 has ☐ cubes.**

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

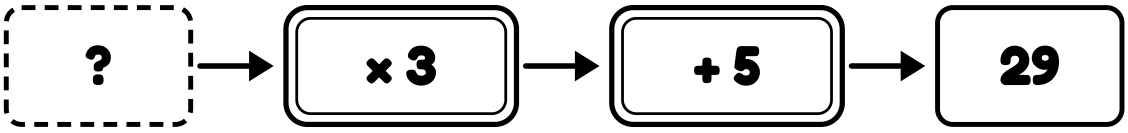
# Learn it: Working backwards

**My goal. I can find a mystery number by working backwards. I undo the last step first. I can use bars for before-and-after stories.**

A number machine changes a number, one step at a time. To find what went IN, work backwards. Undo the LAST step first. Use the opposite: + and − undo each other, and so do × and ÷.

## Worked example

**Ali picks a secret number. He multiplies it by 3, then adds 5. He gets 29. What is his number?**




1. **Undo the last step. It was + 5, so take away 5: 29 − 5 = 24.**
2. **Undo the first step. It was × 3, so divide by 3: 24 ÷ 3 = 8.**
3. **Check going forwards: 8 × 3 = 24, then 24 + 5 = 29.**

**Answer: The secret number is 8.**

## How to work backwards

1. Draw the machine.
2. Start at the OUT. Undo the last step first.
3. Use the opposite step.
4. Check going forwards.

## Opposites

- + 6 is undone by − 6.
- × 4 is undone by ÷ 4.
- Undo in the opposite order.

## Your turn

Answer at the back of the book.

**Try** **Mei picks a secret number. She multiplies it by 5, then takes away 8. She gets 32. What is her number?** ☐

## Start here

Do these first. Take your time.

**1** **What is my number? Write it in the box.**

(a) I add 27. I get 61. ☐

(b) I multiply by 5. I get 35. ☐

(c) I divide by 4. I get 9. ☐

(d) I take away 18. I get 25. ☐

**2** **Circle the step that undoes the first one.**

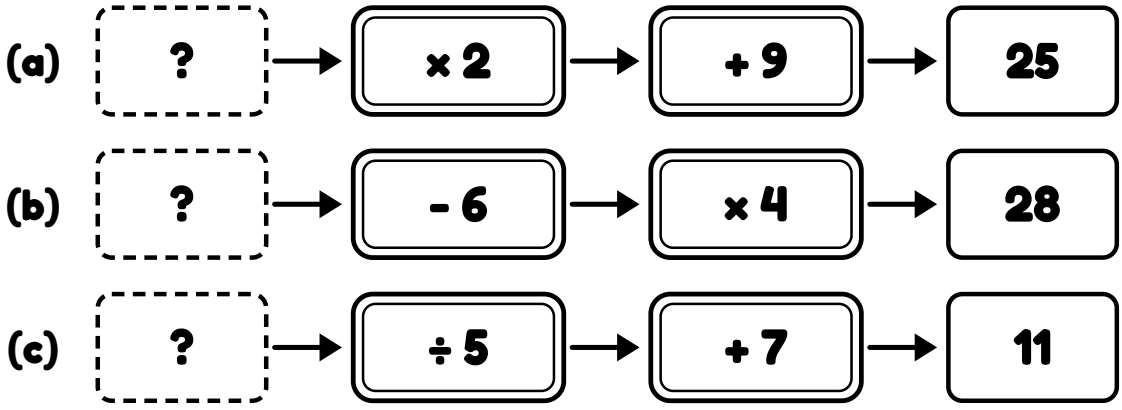
(a) + 7 is undone by + 7 − 7 × 7

(b) × 5 is undone by × 5 ÷ 5 − 5

(c) − 12 is undone by − 12 + 12 ÷ 12

(d) ÷ 3 is undone by ÷ 3 + 3 × 3

**3** **Find the number that went IN. Undo the last step first.**




(a) IN ☐ (b) IN ☐ (c) IN ☐

**4** **Fill in the missing number.**

(a) ☐ ÷ 2 + 7 = 15 (b) 4 × ☐ − 5 = 19

(c) ☐ × 10 + 8 = 78

## Start here

Keep going. Check your work.

**5** **This machine does × 4, then − 1. Fill in the table.**

| IN | 3 | 6 | | |
|---|---|---|---|---|
| OUT | | | 31 | 39 |

(a) Write the 4 missing numbers. Do the OUT of IN 3 and 6 first, then work backwards for 31 and 39.

**6** **A fair machine gives stickers for tokens. Each token gives the same number of stickers, plus a bonus. 3 tokens give 11 stickers. 6 tokens give 17.**

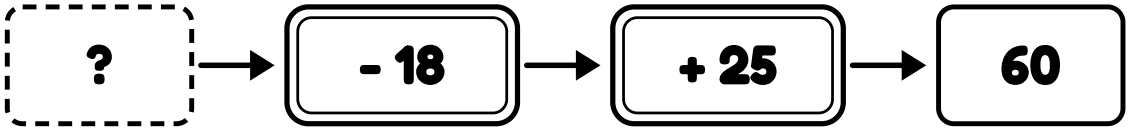
| Tokens | 3 | 6 | 10 |
|---|---|---|---|
| Stickers | 11 | 17 | |

(a) Each token gives 2 3 4 stickers.

(b) The bonus is ☐ stickers.

(c) 10 tokens give ☐ stickers.

**7** **Ali had some marbles. He lost 18 in a game. Then he won 25 marbles. Now he has 60.**




(a) Before he won 25, Ali had ☐ marbles.

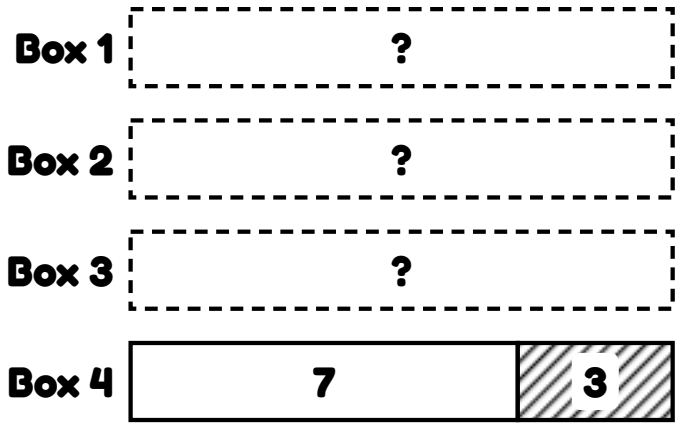
(b) At first Ali had ☐ marbles.

(c) Check: 53 − 18 = ☐, then + 25 = ☐

## Start here

Nearly there. You can do it.

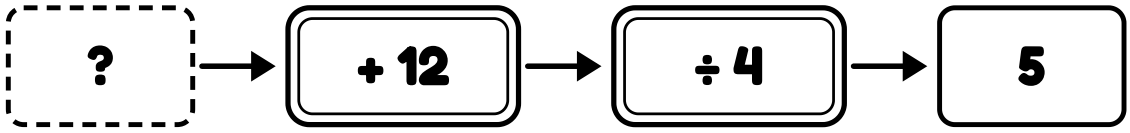
**8** **Kumar puts his stickers into 4 boxes. Every box has the same number. He takes 3 stickers out of box 4. Now box 4 has 7 stickers.**




(a) Each box had ☐ stickers at first.

(b) Kumar has ☐ stickers altogether.

**9** **Mei had some stickers. She bought 12 more. Then she put them equally into 4 albums, 5 in each album.**




(a) After buying 12 more, Mei had ☐ stickers.

(b) At first Mei had ☐ stickers.

**10** **Lily had some money. She bought a pen for $3 and a book for $8. She has $12 left.**

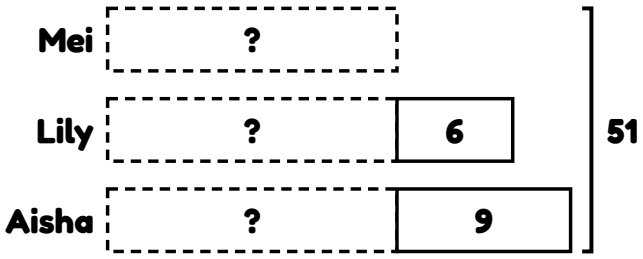
(a) Lily spent $ ☐ altogether.

(b) At first Lily had $ ☐

## Think harder

Hints are upside down. Try first!

**1** **Mei, Lily and Aisha have 51 beads altogether. Lily has 6 more than Mei. Aisha has 9 more than Mei.**

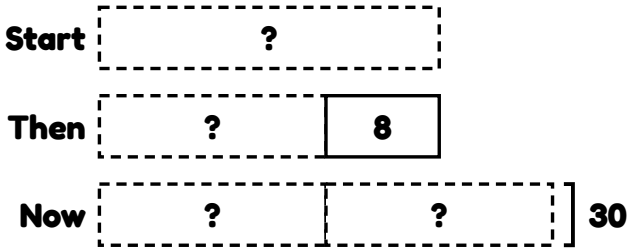



(a) Mei has ☐ beads. (b) Lily has ☐ beads.

(c) Aisha has ☐ beads.

**Hint.** Take off the extra bits. Then share the rest equally.

**2** **Aisha had some stickers. She used 8 on a card. Then she got as many stickers as she had left. Now she has 30.**




(a) After the card, Aisha had ☐ stickers.

(b) At first she had ☐ stickers.

**Hint.** Now she has 2 equal parts. Find one part first.

**3** **A machine does × 4, then + 1. Could it give this OUT? Try IN 1, 2, 3 ... first.**

(a) 29 Yes No (b) 30 Yes No (c) 33 Yes No

(d) 42 Yes No

**Hint.** Write the first OUT numbers. What do you see?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: Letters for numbers

**1** **A letter can stand for a mystery number. Let n be that number. 3 × n + 4 = 25 means: 3 times the number, plus 4, makes 25.**

(a) 3 × n + 4 = 25. n = ☐

(b) 5 × n − 3 = 27. n = ☐

(c) n ÷ 2 + 8 = 13. n = ☐

**Hint.** Work backwards, just like a machine.

**2** **Three jars have 36 beans altogether. 4 beans move from jar A to jar B. Then 6 move from jar B to jar C. Now every jar has the same number.**

(a) Now each jar has ☐ beans.

(b) At first: jar A ☐, jar B ☐, jar C ☐

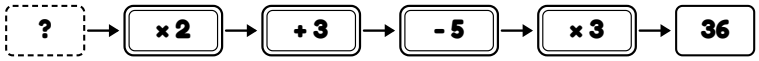
**Hint.** Undo the last move first. Jar C gave nothing, it got 6.

**3** **Make your own riddle. "I think of a number. I … then I … I get …" The mystery number must be 12. Many answers work.**

My riddle: _____

**Hint.** Start at 12 and go forwards.

**4** **Machine A does × 2, then + 3. Machine B does − 5, then × 3. The OUT of A goes into B. The final OUT is 36.**




(a) The OUT of machine A was ☐

(b) The number that went into A was ☐

**Hint.** Undo machine B first. Then undo machine A.

## ✔ Check yourself

No hints. Try each one on your own.

**1** I multiply a number by 4, then add 6. I get 38. My number: ☐

Working: _____

**2** Kumar gave away 14 stickers, then bought 9. Now 31. At first: ☐

Working: _____

**3** Kumar doubles his number, then adds 9. He gets 25. His number: ☐

Working: _____

**4** 5 equal bags. Bag 5 has 4 left after losing 6. Pens in all: ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# Learn it: Money

**My goal.** I can read and write money in dollars and cents. I can pay exactly, find every way, give change and work backwards.

Money is written in dollars and cents. $1 is 100¢, so $3.45 is 345¢. To pay exactly with the fewest pieces, take the greatest note or coin that fits, again and again.

## Worked example

**A cake costs $4.35. Pay the exact amount with as few notes and coins as you can.**




1. **Dollars first: $4 is $2 + $2. That is 2 notes.**
2. **Now the cents: 35¢. The greatest coin that fits is 20¢. 15¢ is left.**
3. **The greatest coin that fits 15¢ is 10¢. Then 5¢ is left: one 5¢ coin.**
4. **Check: 4.00 + 0.20 + 0.10 + 0.05 = 4.35. That is 5 pieces.**

**Answer: $2, $2, 20¢, 10¢ and 5¢. That is 5 pieces.**

## How to pay exactly

1. Split the cost into dollars and cents.
2. Take the greatest note or coin that fits.
3. Repeat with what is left.
4. Add to check. Count the pieces.

## Our money

- Coins: 5¢, 10¢, 20¢, 50¢, $1.
- Notes: $2, $5, $10, $50, $100.
- There is no $20 note.

## Your turn

Answer at the back of the book.

Try **Pay exactly $6.85 with as few notes and coins as you can. How many pieces?** ☐

## Start here

Do these first. Take your time.

**1** **Change dollars and cents.**

(a) $3.45 = ☐ ¢ (b) 275¢ = $ ☐

(c) 6 dollars and 5 cents = $ ☐

(d) 1 dollar and 30 cents = ☐ ¢

**2** **Count the money. How much is there altogether?**




(a) The total is $ ☐

**3** **Pay the exact amount with as few notes and coins as you can. Write them on the lines. Then count the pieces.**




(a) Marker pen: ☐ pieces (b) Kite: ☐ pieces

(c) Backpack: ☐ pieces

Marker pen: _____

Kite: _____

Backpack: _____

## Start here

Keep going. Check your work.

**4** **Find EVERY way to pay exactly $7 using $1 coins, $2 notes and $5 notes. Begin with as many $5 notes as you can. Write each way in a row. Some rows may stay empty.**

| Way | $5 | $2 | $1 | Pieces |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |
| D | | | | |
| E | | | | |
| F | | | | |
| G | | | | |

(a) How many ways are there? ☐

(b) The way with exactly 4 pieces has ☐ $2 notes and ☐ $1 coins.

**5** **Find the change.**

(a) A ruler costs 70¢. Kumar pays with a $1 coin. Change: ☐ ¢

(b) A book costs $4.30. Mei pays $10. Change: $ ☐

(c) A bag costs $37. Ali pays with a $50 note. Change: $ ☐

(d) The fewest notes and coins for the $13 change: ☐ pieces

## Start here

Nearly there. You can do it.

**6** Ravi has $30.




(a) The shoes and the T-shirt cost $ ☐ together.

(b) Can Ravi buy all 3 things? Yes No

(c) He needs $ ☐ more to buy all 3.

**7** Aisha spent $3.50 on lunch and $1.25 on a drink. She has $4.75 left. At first she had $ ☐

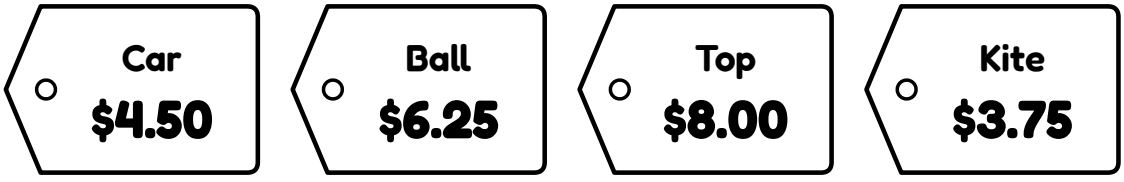
**8** Shop deals.

(a) 4 pencils cost $1. 12 pencils cost $ ☐

(b) 5 stickers cost $2. 20 stickers cost $ ☐

(c) $6 buys ☐ stickers.

**9** Ali buys 2 toys for exactly $12.50. Which 2? ☐ ☐




**10** Mei pays exactly 95¢ with 4 coins. Use 5¢, 10¢, 20¢ or 50¢ coins. Write the coins. ☐ ¢ + ☐ ¢ + ☐ ¢ + ☐ ¢

## Think harder

Hints are upside down. Try first!

**1** **Pay 35¢ with 5¢, 10¢ and 20¢ coins. List EVERY way in the table.**

(a) How many ways are there? ☐

(b) The 5-coin way: ☐ coins of 10¢ and ☐ of 5¢.

| 20¢ | 10¢ | 5¢ |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

**Hint.** Try 1 coin of 20¢ first. Then none. Fill the rest with 10¢ and 5¢.

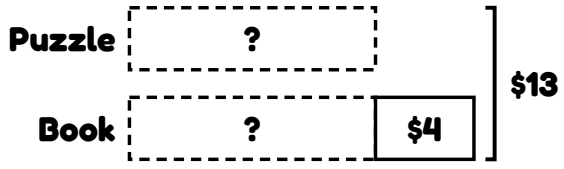
**2** **Lollipops cost 20¢ each. Cookies cost 50¢ each. Siti buys 7 things and spends exactly $2.60. How many lollipops and how many cookies?**

(a) Lollipops: ☐

(b) Cookies: ☐

**Hint.** Start with 7 lollipops. Swap one for a cookie each time.

**3** **A book costs $4 more than a puzzle. Together they cost $13.**




(a) Puzzle: $ ☐ (b) Book: $ ☐

**Hint.** Take away the extra $4. Share what is left equally.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: Halving money

**1** **A shop sign says PAY HALF. You pay half of the amount on the tag.**

(a) A ball has a tag of $9.60. You pay $ ☐

(b) A hat has a tag of $14.50. You pay $ ☐

(c) A kite costs $5.80 with PAY HALF. The tag says $ ☐

**Hint.** Half of dollars first. Then half of cents. Go back with double.

**2** **Mei has 5 coins. Each coin is 20¢ or 50¢.**

(a) How many different amounts could she have? ☐

(b) With exactly 3 coins of 50¢, she has $ ☐

(c) Could she have $1.40? Yes No

**Hint.** Start with 5 coins of 20¢. Swap one for 50¢ each time.

**3** **Make your own coin riddle. Choose 3 coins from 5¢, 10¢, 20¢ and 50¢ (a coin can be used twice). Write how much they are worth, so that only ONE set of 3 coins fits. Many answers are right.**

My riddle: _____

**Hint.** Try all the sets of 3 coins. Does another set make the same total?

**4** **Shop A sells 3 pens for $2. Shop B sells 4 pens for $3. Ben wants 12 pens.**

(a) At shop A, 12 pens cost $ ☐

(b) At shop B, 12 pens cost $ ☐

(c) Which shop is cheaper? A B

**Hint.** 12 pens is how many groups of 3? Of 4?

## ✔ Check yourself

No hints. Try each one on your own.

**1** $4.05 = ☐ ¢

Working: _____

**2** Pay exactly $6.35 using the fewest pieces. Pieces: ☐

Working: _____

**3** A cake costs $7.40. Ravi pays with a $10 note. Change: $ ☐

Working: _____

**4** Ali spent $2.40 on a drink and $3.10 on a snack. He has $4.50 left. At first: $ ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

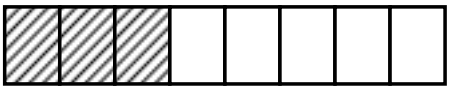
# Learn it: Fractions

**My goal. I can name fractions of a whole. I can compare fractions and add like fractions. I can solve fair-sharing stories with a bar.**

A fraction names equal parts of one whole. The bottom number says how many equal parts the whole is cut into. The top number says how many parts we take.

## Worked example 1

**Kumar cuts a pizza into 8 equal slices. He eats 3 slices. What fraction did he eat? What fraction is left?**

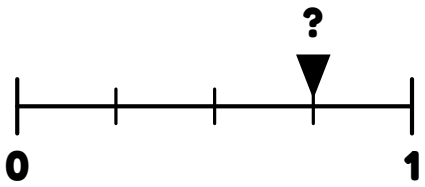


1. **The slices are equal. There are 8 of them: the bottom number is 8.**
2. **He eats 3 slices: the top number is 3. He ate 3/8.**
3. **Slices left: 8 − 3 = 5. So 5/8 is left.**
4. **Check: 3/8 + 5/8 = 8/8 = 1 whole pizza.**

**Answer: He ate 3/8. 5/8 is left.**

## Worked example 2

**The number line goes from 0 to 1 in 4 equal steps. What fraction does the arrow show?**




1. **Each step is 1/4.**
2. **The arrow is 3 steps from 0: 3/4.**

**Answer: The arrow shows 3/4.**

## How to name a fraction

1. Check that the parts are equal.
2. Count all the parts: the bottom number.
3. Count the parts you take: the top number.
4. Same bottom number? Just add or subtract the tops.

## Remember

- More parts means smaller parts: 1/6 is less than 1/4.
- Like fractions: 2/7 + 3/7 = 5/7. The bottom number stays.
- All the parts make 1 whole: 7/7 = 1.

## Your turn

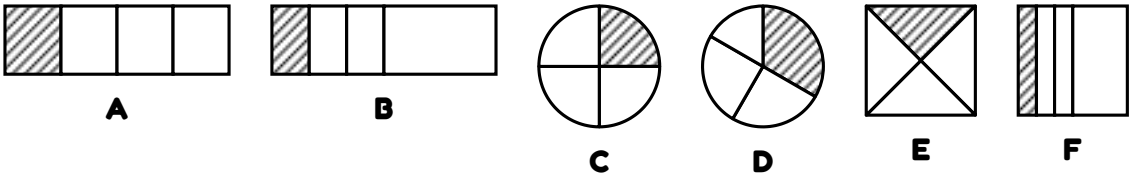
Answer at the back of the book.

**Try** **Mei cuts a cake into 10 equal slices and eats 3. The part of the cake left is** ☐

## Start here

Do these first. Take your time.

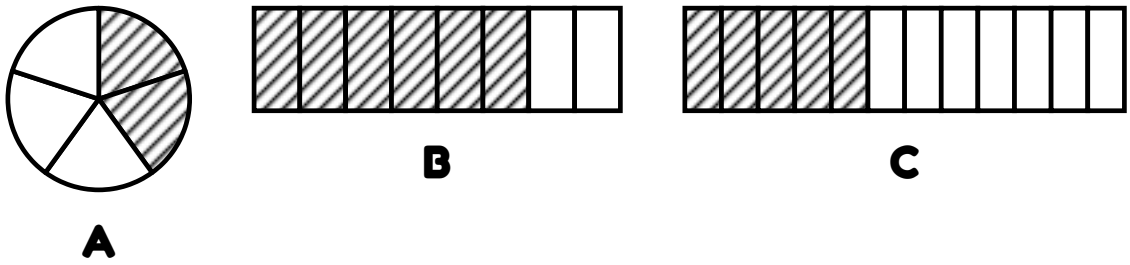
**1** **Each shape is cut into 4 parts. Is each shape cut into quarters (4 EQUAL parts)?**




(a) A Yes No (b) B Yes No (c) C Yes No

(d) D Yes No (e) E Yes No (f) F Yes No

**2** **Write the shaded fraction. Write the fraction not shaded.**




(a) A: shaded ____ not shaded ____

(b) B: shaded ____ not shaded ____

(c) C: shaded ____ not shaded ____

(d) To make 8/12 shaded in C, shade ____ more parts.

**3** **The whole is 1. Fill in the missing number.**

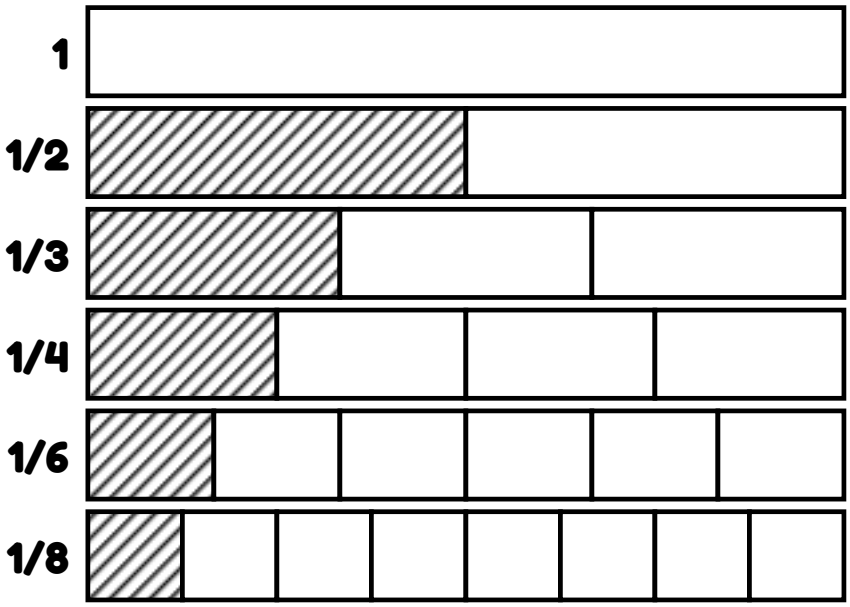
(a) 6/6 = ____ whole (b) 1 whole = 9/____

(c) 4/8 + ____ /8 = 1

## Start here

Keep going. Check your work.

**4** **The fraction wall shows bars of the same length. One part of each bar is shaded.**




(a) Which is greater? 1/3 1/4

(b) Which is smaller? 1/6 1/8

(c) Smallest first: 1/2, 1/6, 1/4, 1/8

_____

**5** **The same bottom number means the same size parts. Circle the greater fraction.**

(a) Circle: 3/10 6/10 (b) Circle: 7/12 5/12

(c) Smallest first: 6/7, 2/7, 4/7

_____

**6** **Add or take away. The bottom number stays the same.**

(a) 3/12 + 6/12 = ☐ (b) 7/8 - 2/8 = ☐

(c) 1 - 3/5 = ☐ (d) 4/9 + ☐ = 1

## Start here

Nearly there. You can do it.

**7** **A pizza is cut into 10 equal slices. Siti eats the slices marked S. Ravi eats the slices marked R.**

| S | S | S | R | R | R | R | | | |
|---|---|---|---|---|---|---|---|---|---|

(a) They eat ☐ of the pizza altogether.

(b) ☐ of the pizza is left.

(c) Who eats more? Siti Ravi

(d) Ravi eats ☐ more than Siti.

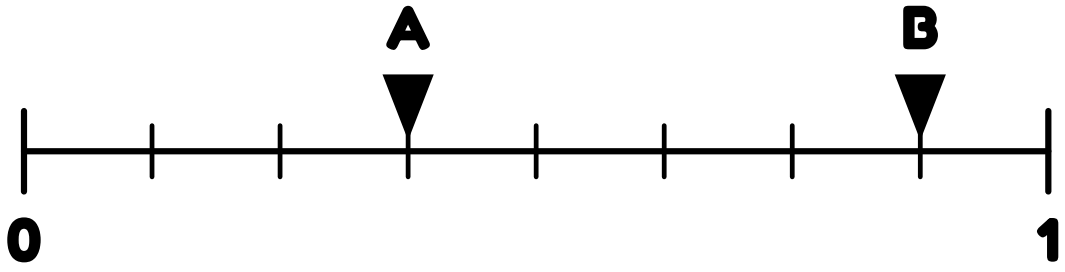
**8** **A cake is shared equally among 5 children.**

(a) Each child gets ☐ of the cake.

(b) Kumar and Mei are 2 of the children. Together they get ☐.

(c) The other children get ☐ together.

**9** **The number line goes from 0 to 1. It is cut into 8 equal steps.**



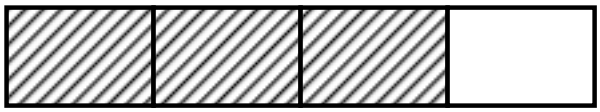

(a) Arrow A shows ☐

(b) Arrow B shows ☐

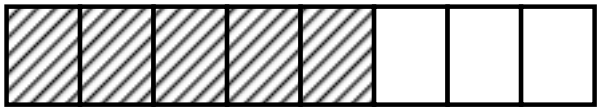
## Think harder

Hints are upside down. Try first!

**1** **Ravi and Siti each have a pizza of the same size. Ravi eats 3/4 of his pizza. Siti eats 5/8 of hers.**



**Ravi: 3/4**



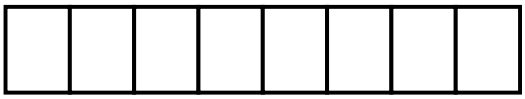
**Siti: 5/8**

(a) Who eats more? Ravi Siti

(b) How much more? ☐

**Hint.** Line up the bars. Cut each quarter in 2 equal parts.

**2** **Mei eats 1/4 of a pizza. Then Ali eats half of what is left.**



**Pizza bar**

(a) Left: ☐ (b) Ali eats: ☐

(c) Now left: ☐

**Hint.** Cut each part in 2 equal parts to make eighths.

**3** **Two fractions have the bottom number 12. Together they make 1 whole. The greater one is 6/12 more.**

(a) Smaller: ☐ (b) Greater: ☐

**Hint.** Take away the extra 6/12. Share what is left.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: Equal fractions

## 1 Equal fractions. Different fractions can cover the same length. 1/2 and 2/4 and 4/8 are equal.

(a) 1/2 = ☐/6 (b) 2/3 = ☐/6 (c) 3/4 = ☐/8

**Hint.** Line up 2 bars of the same length. Cut one into more parts.

## 2 Nine equal pieces of cake are shared by 3 children. The shares must all be different. Each share is 1 piece or more.

(a) How many different ways are there? ☐

(b) In the way where the greatest share is 6/9, the other shares are ☐/9 and ☐/9.

Working: _____

**Hint.** Start with the smallest share, 1 piece. Then try 2.

## 3 Make your own fair-share story. Picture a cake with 8 equal pieces. Three people eat different shares and some is left. Write each share as a fraction. Many answers are right.

My story: _____

Check: shares and what is left make 1 whole: _____

**Hint.** Add all the shares and what is left. Do you get 8/8?

## 4 My top number is 1. I am less than 1/3 and more than 1/9. Which fractions fit? Find ALL of them.

(a) I could be: _____

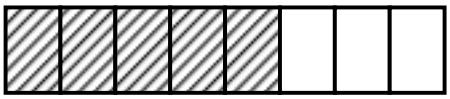
(b) How many answers are there? ☐

**Hint.** Greater bottom number, smaller part. Try 1/4, 1/5 …

## ✔ Check yourself

No hints. Try each one on your own.

**1** What fraction is shaded? ☐



**2** Which is greater? 1/5 1/9

Working: _____

**3** 2/7 + 4/7 = ☐

Working: _____

**4** A pizza has 12 slices. Ali eats 5/12 and Mei eats 3/12. What is left, in twelfths? ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

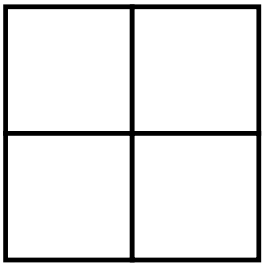
# Learn it: 2D and 3D shapes

**My goal.** I can name half circles and quarter circles. I can count shapes and cubes in order. I can use layers to count a stack.

Shapes can be put together and taken apart. A half circle has 1 straight side. A quarter circle has 2 straight sides. To count shapes, work in order and count one kind at a time.

## Worked example 1

**Count every square in this grid, big and small. It has 2 rows and 2 columns.**

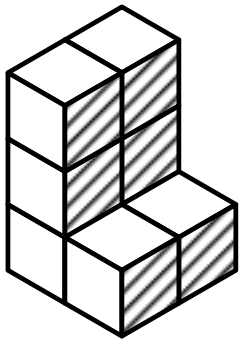


1. Count the small squares first: 4.
2. Look for larger squares. One square is 2 by 2: the whole grid. That is 1.
3. Add: 4 + 1 = 5 squares.

**Answer: 5 squares in all.**

## Worked example 2

**Count the cubes in this stack. It is 2 cubes deep. The back is like the front.**



1. Count one row: 3 + 1 = 4 cubes.
2. The 2 rows are the same: 4 + 4 = 8 cubes.

**Answer: The stack has 8 cubes.**

## How to count shapes

1. Count the smallest shapes first.
2. Then count the next size up.
3. Keep going to the largest.
4. Add the counts.

## Solids

- Cube: 6 flat square faces.
- Cylinder: 2 flat circles and 1 curved face.
- Cone: 1 flat circle and 1 curved face.
- Sphere: no flat face.
- Stacks: count row by row.

## Your turn

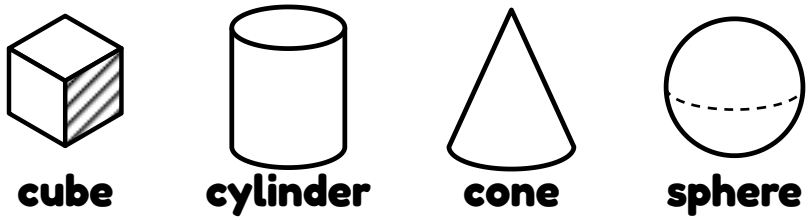
Answer at the back of the book.

Try **A grid has 2 rows and 3 columns. How many squares are there, big and small?**

## Start here

Do these first. Take your time.

**1** **Look at the solids. How many flat faces does each solid have?**

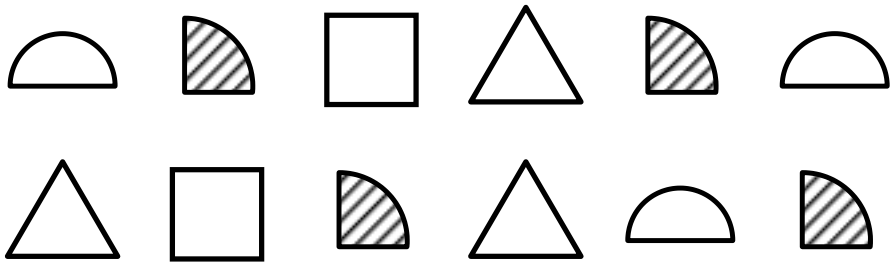


cube cylinder cone sphere

(a) Cube: ☐ (b) Cylinder: ☐ (c) Cone: ☐

(d) Sphere: ☐

**2** **Count each kind of piece.**



(a) Half circles: ☐ (b) Quarter circles: ☐

(c) Squares: ☐ (d) Triangles: ☐

**3** **All the circles are the same size.**

(a) ☐ quarter circles make 1 half circle.

(b) 3 half circles make ☐ quarter circles.

(c) 12 quarter circles make ☐ whole circles.

## Start here

Keep going. Check your work.

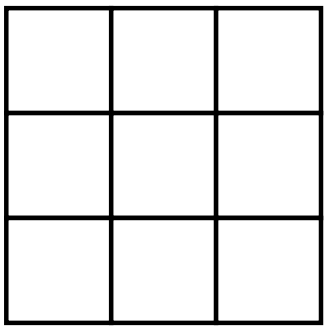
**4** **Count every square of every size in this grid. Start with the smallest.**

(a) 1 by 1 squares: ☐

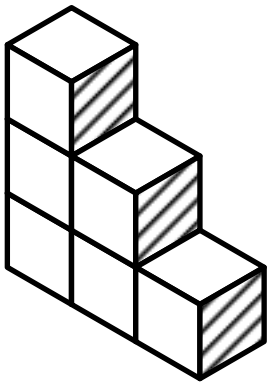
(b) 2 by 2 squares: ☐

(c) 3 by 3 squares: ☐

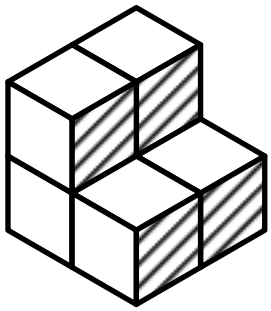
(d) All the squares: ☐



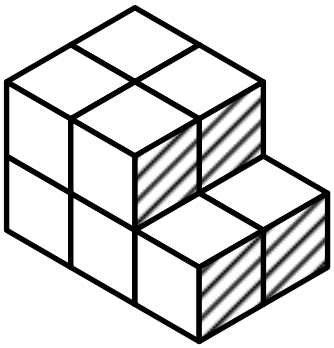
**5** **Count the cubes in each stack. Every cube sits on the table or on another cube. Stack A is 1 cube deep. Stacks B and C are 2 cubes deep. At the back they are just like at the front.**



A



B



C

(a) A: ☐ cubes (b) B: ☐ cubes (c) C: ☐ cubes

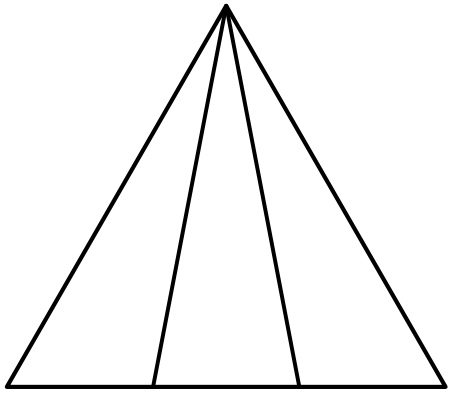
**6** **Count the triangles of all sizes. Start with the smallest.**

(a) Small triangles: ☐

(b) Triangles made of 2 small ones: ☐

(c) The whole triangle: ☐

(d) All the triangles: ☐



## Start here

Nearly there. You can do it.

**7** **Which solid am I? Circle it.**

(a) I have 6 square faces. cube cone sphere

(b) I have 1 flat circle and 1 curved face. cone cylinder cube

(c) I roll and have no flat face. sphere cube cylinder

(d) I have 2 flat circles. cone cylinder sphere

**8** **Put the pieces together. Circle the shape you make.**

(a) 2 half circles make a circle square triangle

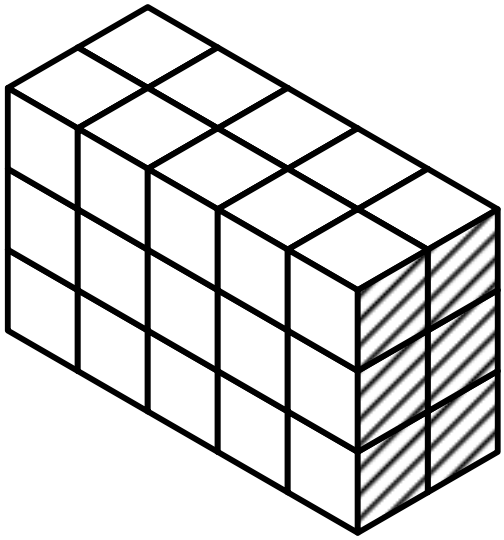
(b) 2 same squares make a rectangle circle triangle

(c) 4 quarter circles make a circle half circle square

**9** **Ali builds a stack of cubes. The bottom layer is 5 cubes long and 2 cubes wide. The stack has 3 layers that are all the same.**

(a) Each layer has ☐ cubes.

(b) The stack has ☐ cubes.



**10** **Name the flat faces.**

(a) Each face of a cube is a square circle triangle

(b) Each flat face of a cylinder is a circle square triangle

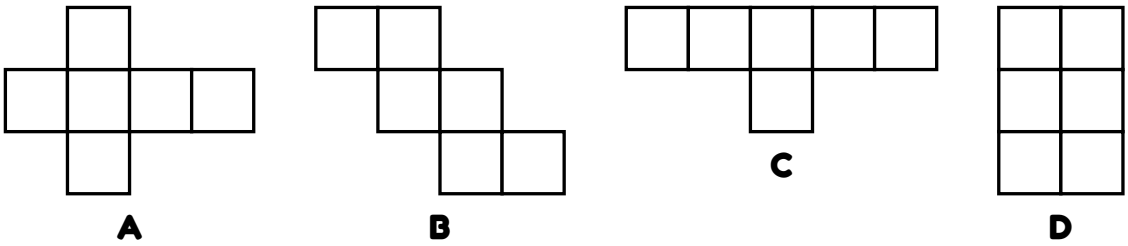
(c) The flat face of a cone is a triangle circle square

## Think harder

Hints are upside down. Try first!

★ Beyond Level 2: Cube nets

**1** **A net folds up into a solid. Which nets fold into a cube? Each square must be a new face.**

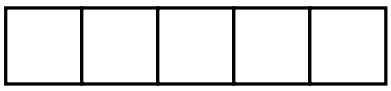



**(a)** A Yes No **(b)** B Yes No **(c)** C Yes No

**(d)** D Yes No

**Hint.** Fold in your head. Where does each square land?

**2** **A strip has 5 stamps in a row. Mei tears off 1 or more stamps that touch.**



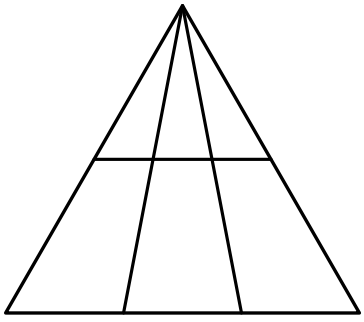
**(a)** Pieces of 1 stamp: ☐ **(b)** Pieces of 2 stamps: ☐

**(c)** Pieces of 3 stamps: ☐

**(d)** Pieces of 4 or 5 stamps: ☐ All the pieces: ☐

**Hint.** A piece of 2 stamps can start at stamp 1, 2, 3 or 4.

**3** **A big triangle has 1 line across. It has 2 lines from its top corner.**



**(a)** Sitting on the base: ☐

**(b)** Sitting on the middle line: ☐

**(c)** All the triangles: ☐

**Hint.** Count the triangles that sit on each line separately.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: Triangle numbers

**1** **A shop stacks cans in a triangle. The top row has 1 can. Each row has 1 more can.**

**(a)** 6 rows: the bottom row has ☐ cans.

**(b)** 6 rows: ☐ cans in all.

**(c)** 8 rows: ☐ cans in all.

**Hint.** Add the rows: 1 + 2 + 3 …

★ Beyond Level 2: Painted cubes

**2** **Small cubes make a big cube that is 4 long, 4 wide and 4 high. The outside is painted. Count the small cubes with each number of painted faces.**

**(a)** 3 faces painted: ☐ **(b)** 2 faces painted: ☐

**(c)** 1 face painted: ☐ **(d)** No paint: ☐

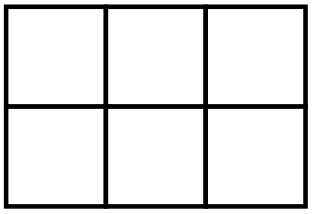
**Hint.** Corners, edges, faces, inside. Check they make 64.

**3** **Design a stack with exactly 24 cubes. Write the cubes in each layer. Many answers are right.**

Stack 1: _____

**Hint.** Each layer needs a layer under it.

**4** **Count the rectangles in a grid with 2 rows and 3 columns. A square counts too.** ☐



**Hint.** Sort them by size: 1 by 1, 1 by 2, 1 by 3, 2 by 1, 2 by 2, 2 by 3.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Which solid has 1 flat face and 1 curved face?**

**cube** **cone** **cylinder**

Working: _____

**2** **All the circles are the same size. 3 half circles make ☐ quarter circles.**

Working: _____

**3** **A stack has 2 layers. Each layer is 4 cubes by 3 cubes. How many cubes? ☐**

Working: _____

**4** **A grid has 3 rows and 2 columns. Squares, big and small: ☐**

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

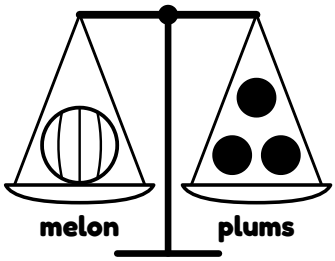
# Learn it: Length, mass, volume and time

**My goal.** I can solve balance puzzles and pouring puzzles. I can change units. I can tell the time to 5 minutes and find how long.

A level balance has the same mass on both sides, so you can swap things. The short hand shows hours. The long hand shows minutes.

## Worked example 1

**The scale is level. 1 melon balances 3 plums. 1 plum has a mass of 50 g. Find the mass of the melon.**




1. The melon is as heavy as 3 plums.
2. 3 plums: 3 × 50 g = 150 g.
3. So the melon is 150 g.

**Answer: The melon is 150 g.**

## Worked example 2

**A concert starts at 7:55 pm and lasts 30 minutes. When does it end?**




1. 7:55 to 8:00 is 5 minutes.
2. 30 − 5 = 25 minutes more.
3. So it ends at 8:25 pm.

**Answer: It ends at 8:25 pm.**

## How to start

1. Balance: both sides are equal. Swap things.
2. Time: count on to the next hour first.
3. Jugs: write both amounts after each pour.
4. Check your answer.

## Units and time

- am: midnight to noon. pm: noon to midnight.
- 1 kg = 1000 g. 1 m = 100 cm.
- Long hand: count in 5s.
- Short hand: read the smaller number.
- 60 minutes = 1 hour.

## Your turn

Answer at the back of the book.

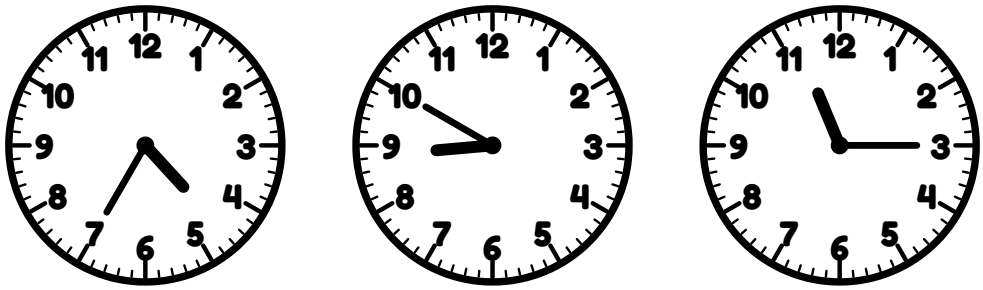
Try **One tin balances 4 pens. One pen has a mass of 25 g. Find the mass of the tin.**

☐ g

## Start here

Do these first. Take your time.

**1** **Read the clocks. Write each time.**



A B C

(a) A: ☐ : ☐ (b) B: ☐ : ☐

(c) C: ☐ : ☐

**2** **Write am or pm.**

(a) Mei eats breakfast at 7:30 am pm

(b) School ends at 1:30 am pm

(c) Ali goes to bed at 9:15 am pm

(d) The sun rises at 6:40 am pm

**3** **Find the end or start time.**

(a) A film begins at 3:40 pm and runs for 35 minutes. It finishes at ☐ : ☐ pm.

(b) A class starts at 9:50 am and ends at 10:30 am. It lasts ☐ minutes.

(c) A game ends at 5:05 pm. It lasted 25 minutes. It started at ☐ : ☐ pm.

## Start here

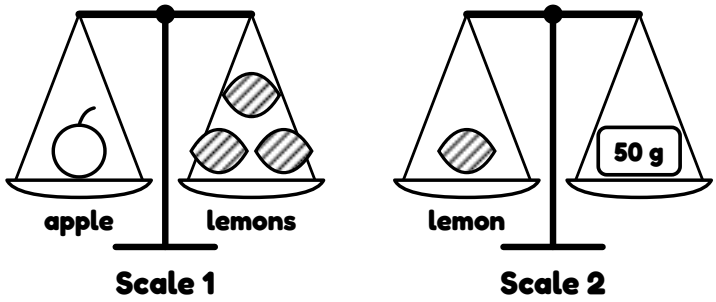
Keep going. Check your work.

**4** **A trip starts at 9:35 am and ends at 11:05 am. How long is the trip?**

(a) The trip is ☐ hour and ☐ minutes long.

Draw a timeline:

**5** **All the scales are level.**



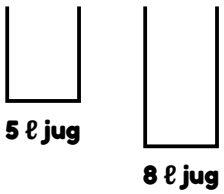

(a) 1 lemon is ☐ g (b) 1 apple is ☐ g

**6** **A 5 ℓ jug and an 8 ℓ jug have no marks. Both are empty.**

(a) Fill the 8 ℓ jug. Pour into the 5 ℓ jug until it is full.

The 8 ℓ jug holds ☐ ℓ.

(b) Empty the 5 ℓ jug. Pour the 3 ℓ in. It needs ☐ ℓ more.




(c) Fill the 8 ℓ jug. Pour into the 5 ℓ jug until it is full.

The 8 ℓ jug holds ☐ ℓ.

**7** **Circle the better unit.**

(a) The mass of an egg g kg

(b) The length of a pencil cm m

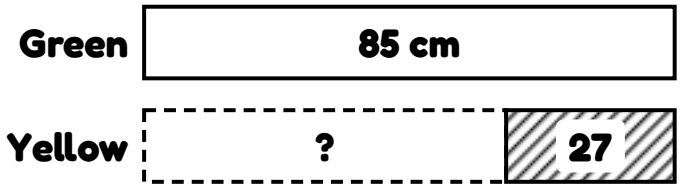
(c) The mass of a school bag g kg

(d) Water in a jug ℓ cm

## Start here

Nearly there. You can do it.

**8** Lily has a green ribbon 85 cm long. Her yellow ribbon is 27 cm shorter.




(a) The yellow ribbon is ☐ cm long.

(b) Lily joins them end to end with an overlap of 4 cm. The long ribbon is ☐ cm.

(c) She cuts 3 pieces of 20 cm from the green ribbon. Green ribbon left: ☐ cm

**9** Change the units.

(a) 3 kg = ☐ g (b) 2500 g = ☐ kg ☐ g

(c) A bag has 2 kg of rice. Mei uses 750 g. Rice left: ☐ g

**10** Change the units.

(a) 2 m 35 cm = ☐ cm (b) 180 cm = ☐ m ☐ cm

(c) A rope is 5 m. Ali cuts off 130 cm. Rope left: ☐ cm

**11** A bucket holds 12 ℓ. It has 4 ℓ in it. Mei pours in 2 ℓ each time.

(a) How many pours fill the bucket? ☐

(b) After 3 pours the bucket holds ☐ ℓ.

## Think harder

Hints are upside down. Try first!

**1** **4 plums balance 1 pear. A pear and 2 plums balance 600 g.**

(a) One plum is ☐ g.

(b) The pear is ☐ g.

**Hint.** Swap the pear for 4 plums. How many plums are on that side?

**2** **A 5 ℓ jug and a 7 ℓ jug have no marks. Get exactly 4 ℓ. Three steps are done.**

| Step | What I do | 5 ℓ jug | 7 ℓ jug |
|---|---|---|---|
| Start | Both jugs empty | 0 | 0 |
| 1 | Fill the 7 ℓ jug | 0 | 7 |
| 2 | Pour into the 5 ℓ jug | 5 | 2 |
| 3 | Empty the 5 ℓ jug | 0 | 2 |
| 4 | | | |
| 5 | | | |
| 6 | | | |

(a) After step 5 the jugs hold ☐ ℓ and ☐ ℓ.

(b) At the end the 7 ℓ jug holds ☐ ℓ.

**Hint.** Pour the 2 ℓ into the 5 ℓ jug. Then fill the 7 ℓ jug.

**3** **A museum tour has 4 stops. Each stop takes 20 minutes. There is a 5-minute walk between stops. The tour starts at 9:15 am.**

(a) The tour takes ☐ minutes.

(b) It ends at ☐ : ☐ am.

**Hint.** 4 stops, but how many walks?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: 24-hour clock

## 1 The 24-hour clock. After noon add 12 to the hour: 1:30 pm is 13:30.

(a) 3:45 pm = ☐ : 45 (b) 18:10 = ☐ : 10 pm

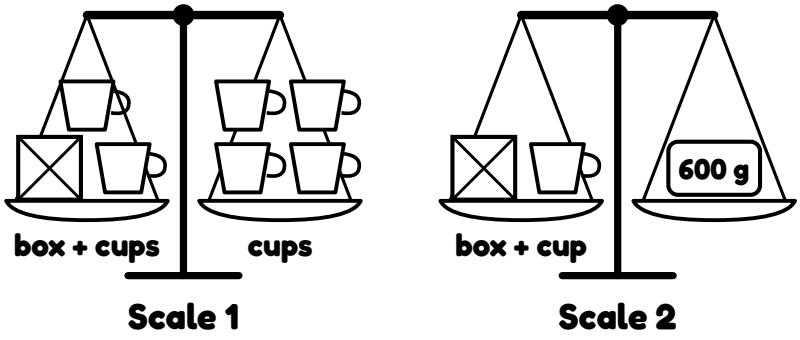
(c) 20:05 = ☐ : 05 pm

**Hint.** After noon: add 12 to the hour. Going back: take 12 away.

## 2 Scale 1: 1 box and 2 cups balance 4 cups. Scale 2: 1 box and 1 cup balance 600 g.

(a) One cup is ☐ g.

(b) One box is ☐ g.




**Hint.** Take 2 cups off both sides of scale 1.

## 3 Make your own balance puzzle with apples and lemons. One apple must be 200 g. Many answers are right.

My scales: _____

**Hint.** Start from 200 g. What balances it?

## 4 A clock gains 3 minutes every hour. It is set right at 7:00 am.

(a) At 11:00 am the clock shows ☐ : ☐ am.

(b) It shows 12:15 pm. The right time is ☐ : ☐ .

**Hint.** How many hours have passed? 3 minutes each hour.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **A show begins at 6:50 pm and runs for 35 minutes. It finishes at** ☐ **:** ☐ **pm.**

Working: _____

**2** **2 kg 300 g =** ☐ **g**

Working: _____

**3** **1 melon balances 3 pears. 1 pear is 200 g. The melon is** ☐ **g.**

Working: _____

**4** **A rope is 3 m long. Ali cuts off 140 cm. Rope left:** ☐ **cm**

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# Learn it: Mixed puzzles

**My goal.** I can choose a way to start a puzzle. I can read picture graphs and logic grids. I can check my answer against every clue.

A puzzle is easier when you know how to start. Draw it. List it in order. Make a table. Then check every clue.

## Worked example

**Lily has 7 coins. Each coin is 10¢ or 20¢. They are worth $1.10 altogether. How many of each coin?**

| 10¢ coins | 7 | 6 | 5 | 4 | 3 |
|---|---|---|---|---|---|
| 20¢ coins | 0 | 1 | 2 | 3 | 4 |
| Total | 70¢ | 80¢ | 90¢ | 100¢ | 110¢ |

1. **Start with 7 coins of 10¢. That is 70¢.**
2. **Swap one 10¢ for a 20¢. The total goes up by 10¢ each time.**
3. **Stop when the total is 110¢: 3 coins of 10¢ and 4 coins of 20¢.**
4. **Check: 30¢ + 80¢ = 110¢ and 3 + 4 = 7 coins.**

**Answer: 3 coins of 10¢ and 4 coins of 20¢.**

## Ways to start

1. Draw it: a picture or a bar.
2. List it: write every case in order.
3. Make a table: try 1, 2, 3 ...
4. Check against every clue.

## Tips

- Graph: read the key first.
- Logic grid: tick one box. Cross out the rest of its row and column.
- List in order so no case is missed.

## Your turn

Answer at the back of the book.

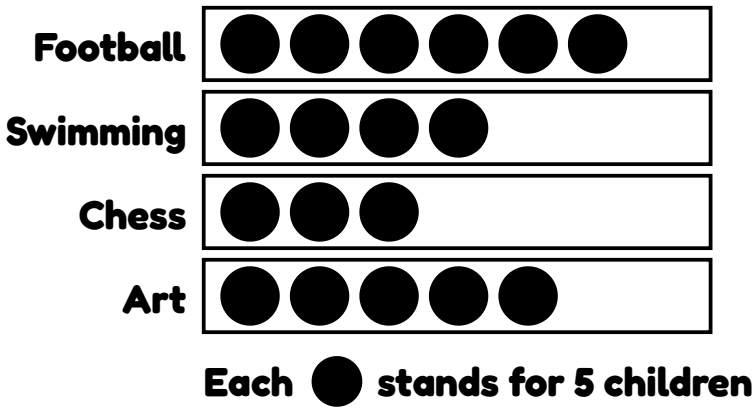
**Try** **The key of a graph says each picture is 5 children. Maths has 6 pictures.**

**How many chose maths?** ☐

## Start here

Do these first. Take your time.

**1** **Read the picture graph. Look at the key first.**

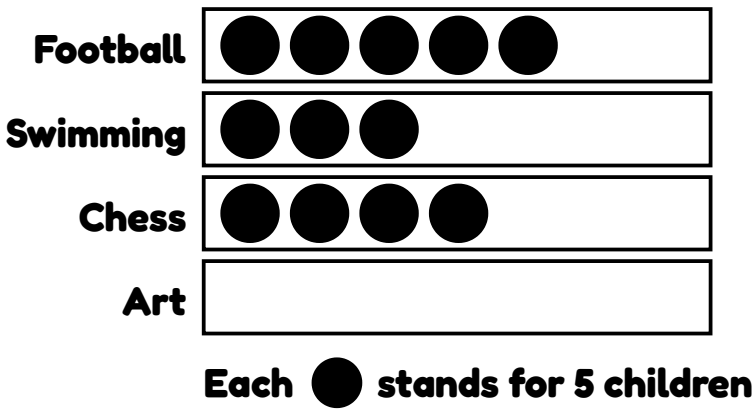



(a) Chess: ☐ children

(b) More children chose football than swimming: ☐

(c) Children altogether: ☐

**2** **75 children chose a club. The Art row is not drawn yet.**




(a) Art needs ☐ pictures. (b) ☐ children chose Art.

**3** **I am a number between 40 and 60. I am in the 5 times table. My digits add up to 10. What number am I?** ☐

## Start here

Keep going. Check your work.

**4** **Ali counts the pets in his street. He makes a tally chart. A group of 5 has 4 strokes and a slash.**

Cats 𝍸 ||

Dogs 𝍸 𝍸 ||

Fish ||||

(a) Dogs: ☐ (b) More dogs than cats: ☐

(c) All the pets: ☐

**5** **The cards 7, 0 and 5 can build 3-digit numbers. Use each card once.**

| 7 | 0 | 5 |
|---|---|---|

(a) List every number in order:

_____

(b) The greatest is ☐ and the smallest is ☐.

(c) How many are between 600 and 800? ☐

**6** **Ben has 6 coins. Each coin is 10¢ or 20¢. They are worth $1.10 altogether. Fill in the table.**

| 10¢ coins | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|
| 20¢ coins | 0 | 1 | 2 | 3 | 4 | 5 |
| Total | 60¢ | 70¢ | | | | |

(a) Write the 4 missing totals:

_____

(b) Ben has ☐ coins of 10¢ and ☐ coins of 20¢.

## Start here

Nearly there. You can do it.

**7** **Ali, Mei and Ravi each have 1 fruit: an apple, a banana or a mango. Mei has the mango. Ali does not have the apple.**

(a) Ali has the ☐

(b) Mei has the ☐

(c) Ravi has the ☐

| | apple | banana | mango |
|---|---|---|---|
| Ali | | | |
| Mei | | | |
| Ravi | | | |

**8** **Mei cuts a pizza into 6 equal slices. She eats 1/6 and Ali eats 2/6. Each slice is 100 g. Write fractions in sixths.**

(a) They eat ☐ of the pizza.

(b) ☐ of the pizza is left.

(c) The mass of the pizza left is ☐ g.

**9** **Aisha buys 3 pens. Each pen costs $1.20. She pays with a $5 note.**

(a) The pens cost $ ☐

(b) Her change is $ ☐

**10** **A hall has 4 rows of 10 chairs. 28 children sit down.**

(a) The hall has ☐ chairs.

(b) ☐ chairs are empty.

## Think harder

Hints are upside down. Try first!

**1** **Ali, Mei, Siti and Ben like swimming, chess, art or tennis, one each. Mei likes chess. Ali does not like chess or art. Siti does not like swimming or tennis. Ben does not like tennis.**

(a) Ali: ☐

(b) Mei: ☐

(c) Siti: ☐

(d) Ben: ☐

| | swim | chess | art | tennis |
|---|---|---|---|---|
| Ali | | | | |
| Mei | | | | |
| Siti | | | | |
| Ben | | | | |

**Hint.** Tick Mei and chess. Cross out the rest of its row and column.

**2** **The key was lost. Apples have 7 pictures and pears have 3. 20 more children chose apples.**

Apples ●●●●●●●

Pears ●●●

**Each ● stands for ?**

(a) Each picture stands for ☐ children.

(b) Apples: ☐ children (c) Pears: ☐ children

**Hint.** 4 more pictures is 20 children. How many in 1 picture?

**3** **Pump A adds 2 l in minute 1, 4 l in minute 2, 6 l in minute 3, and so on. Pump B adds 5 l every minute. Compare the totals in the tank.**

(a) Pump A first adds more than pump B by minute ☐.

(b) Then pump A has ☐ l more.

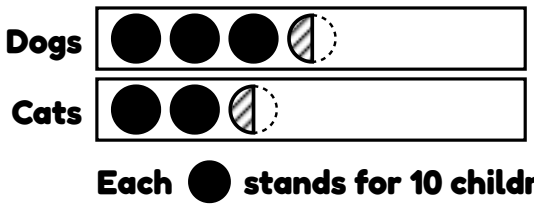
**Hint.** Add each minute to the total of the minute before.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 2: Half pictures

## 1 Half a picture. Each picture stands for 10 children, so half a picture stands for 5 children.




(a) Dogs: ☐ children (b) Cats: ☐ children

(c) More dogs than cats: ☐

**Hint.** 3 pictures is 30. Add the half picture.

## 2 Ali has one 5¢ coin, one 10¢ coin and two 20¢ coins. He pays with one or more of them.

(a) How many different amounts can he pay? ☐

(b) Can he pay 60¢? Yes No

**Hint.** List the amounts in order: 5¢, 10¢, 15¢ ...

## 3 Make your own logic puzzle. Three children have a cat, a dog and a fish. Write 2 clues so only ONE answer fits. Many answers are right.

My clues: _____

**Hint.** Pick the answer first. Then write clues that rule out the rest.

## 4 Ali buys 8 stickers. Small stickers cost 30¢. Large stickers cost 50¢. He pays $3.40 altogether.

(a) He buys ☐ small stickers and ☐ large stickers.

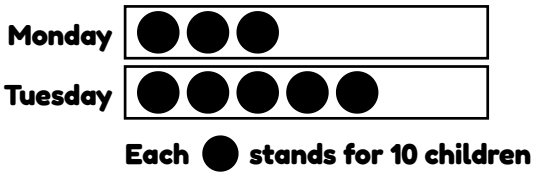
(b) Could 8 stickers cost $3.30? Yes No

**Hint.** Start with 8 small stickers. Each swap adds 20¢.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Each picture stands for 10 children. How many children chose Tuesday?** ☐




**2** **Use the digit cards 1, 0 and 4 once each. How many 3-digit numbers can you make?** ☐

Working: _____

**3** **Ali, Ben and Mei have red, blue and green. Mei has blue. Ali does not have red. Ben has** ☐

Working: _____

**4** **A pizza has 12 slices. 5/12 is eaten. Left, in twelfths:** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# My progress

Tick each box when you finish. Write your Check yourself score.

| Unit | Topic | Learn it | Start here | Think harder | Go further | Score | Date |
|---|---|---|---|---|---|---|---|
| 1 | Digit detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 2 | Missing digit mysteries | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 3 | Legs, wheels and arrays | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 4 | Pattern machines | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 5 | Mystery number machines | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 6 | Market day | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 7 | Fair shares | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 8 | Shape builders | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 9 | Measure masters | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 10 | Puzzle day | ☐ | ☐ | ☐ | ☐ | / 4 | |

## Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____




# Certificate of Completion

## Well done,

**You finished the Math Inquirers Extension Workbook, Level 2.**

Date

Signed by a grown-up

# Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

# Unit 1 · Digit detectives

**Your turn**
page 6

**Answer: 4**
**Why:** List in order: 309, 390, 903, 930. A number cannot start with 0, so 039 and 093 do not count.

**START HERE**

**1**
page 7

**Answer: (a) 356 (b) 804 (c) 9 (d) 30**
**Why:** (a) 300 + 50 + 6 = 356. (b) 800 + 0 + 4 = 804. (c) 590 is 500 + 90, and 90 is 9 tens. (d) The 3 is in the tens place, so it is worth 30.

**2**
page 7

**Answer: 424**
**Why:** 3 flats are 300, 12 rods are 120 and 4 ones are 4. 300 + 120 + 4 = 424. 10 of the rods make another flat, so this is also 4 flats, 2 rods and 4 ones.

**3**
page 7

**Answer: (a) < (b) > (c) > (d) =**
**Why:** (a) The hundreds are the same; 4 tens is less than 5 tens, so 745 < 754. (b) 3 hundreds is more than 2 hundreds, so 306 > 296. (c) 5 hundreds 2 tens is 520, and 520 > 502. (d) 88 tens is 880, so they are equal.

**4**
page 7

**Answer: 85, 508, 580, 805**
**Why:** 85 has only 2 digits, so it is the smallest. 508, 580 and 805 are 3-digit numbers. Compare the hundreds, then the tens: 508 < 580 < 805.

**5**
page 8

**Answer: 158, 185, 518, 581, 815, 851**
**Why:** Start with 1: 158 and 185. Start with 5: 518 and 581. Start with 8: 815 and 851. 3 cards x 2 = 6 numbers.

**6**
page 8

**Answer: (a) 851 and 158 (b) 581 (c) 4**
**Why:** (a) The greatest digit first gives 851; the smallest first gives 158. (b) 581 is 19 from 600. (c) Odd numbers end in 1, 3, 5, 7 or 9: 185, 581, 815, 851.

**7**
page 8

**Answer: (a) 406, 460, 604, 640 (b) 4**
**Why:** 0 cannot be the hundreds digit, so only 6 or 4 can start. Start with 4: 406 and 460. Start with 6: 604 and 640. That is 4 numbers.

**8**
page 8

**Answer: (a) 653 (b) 305**
**Why:** (a) The tens and ones digits add up to 14 - 6 = 8. The tens digit is 2 more than the ones digit: 5 and 3. The number is 653. (b) Odd numbers between 300 and 310: 301, 303, 305, 307, 309. Only 3 + 0 + 5 = 8, so 305.

**9**
page 9

**Answer: (a) 340 (b) 370 (c) 30**
**Why:** Each mark is 10 more. (a) A is 4 marks after 300: 340. (b) B is 7 marks after 300: 370. (c) 370 - 340 = 30.

**10**
page 9

**Answer: (a) 573, 581, 592, 593 (left to right, top to bottom) (b) 635, 644, 646, 655 (top, left, right, bottom) (c) 642**
**Why:** Right is 1 more, down is 10 more. (a) 573, 581, 592, 593. (b) Above 645 is 635, left is 644, right is 646, below is 655. (c) 3 squares down is 30 more: 612 + 30 = 642.

**11** page 9

**Answer: (a) True (b) False (c) False (d) True**
**Why:** (a) 50 + 12 = 62, true. (b) 7 hundreds is 700, which is 70 tens, so false. (c) One 9 is worth 900 and the other is worth 9, so false. (d) 10 hundreds make 1000, true.

## THINK HARDER

**1** page 10

**Answer: (a) 12 (b) 6**
**Why:** (a) Even numbers end in 4 or 6. Ones digit 4: 2 of 1, 6, 9 fill the other places: 3 x 2 = 6 numbers. Ones digit 6: 6 numbers. 6 + 6 = 12. (b) Ending in 4: 614, 694, 914, 964. Ending in 6: 916, 946. That is 6 numbers.

**2** page 10

**Answer: (a) 209, 407, 605, 803 (b) 407**
**Why:** (a) The hundreds and ones digits add up to 11 and the ones digit is odd: (2, 9), (4, 7), (6, 5), (8, 3). (10, 1) is not allowed. The numbers are 209, 407, 605 and 803. (b) 407 is 93 from 500 and 605 is 105 from 500.

**3** page 10

**Answer: (a) 20 (b) 40**
**Why:** (a) 7s in the ones place: 7, 17, ... 97 is 10 numbers. 7s in the tens place: 70 to 79 is 10 numbers. 77 has two 7s. 10 + 10 = 20. (b) 101 to 200 is the same pattern, so 20 more: 20 + 20 = 40.

## GO FURTHER

**1** page 11

**Answer: (a) 4207 (b) 3650 (c) 2500**
**Why:** (a) 4000 + 200 + 0 + 7 = 4207. (b) Adding 1000 adds 1 to the thousands digit: 2650 becomes 3650. (c) 20 hundreds is 2000 and 5 hundreds is 500, so 2500.

**2** page 11

**Answer: 90**
**Why:** The hundreds digit can be 1 to 9: 9 choices. The tens digit can be 0 to 9: 10 choices. The ones digit must be the same as the hundreds digit. 9 x 10 = 90.

**3** page 11

**Answer: 255**
**Why:** The hundreds and ones digits add up to 12 - 5 = 7. Try 156: swapping gives 651, which is 495 more. Try 255: swapping gives 552, which is 297 more. So the number is 255.

**4** page 11

**Answer: Open. Example: "I am a 3-digit number. I am between 300 and 400. My digits add up to 12. My ones digit is 2." Check that no other number fits all 3 clues.**
**Why:** A good riddle has exactly one answer. Example: between 300 and 400 leaves 100 numbers. Digit sum 12 leaves 309, 318, 327 ... Ones digit 2 leaves only 372. Test yours with a friend.

## CHECK YOURSELF

**1** page 12

**Answer: 830**
**Why:** 13 tens is 130. 700 + 130 = 830.

**2** page 12
**Answer: 850**
**Why:** An even number ends in 0 or 8. To be greatest, try 8 first: 850 ends in 0, so it is even. 850 is greater than 580 and 508.

**3** page 12
**Answer: 282, 272**
**Why:** Take away 10 each time. 292 - 10 = 282 and 282 - 10 = 272.

**4** page 12
**Answer: 6**
**Why:** The odd numbers are 531, 533, 535, 537, 539 and 541. That is 6 numbers.

## Unit 2 · Missing digit mysteries

**Your turn** page 13
**Answer: 5**
**Why:** Ones: 4 + 7 = 11, write 1 and carry 1. Tens: 5 + 3 + 1 = 9, so the missing digit is 5. Check: 554 + 337 = 891.

**START HERE**

**1** page 14
**Answer: (a) 4 and 9 (b) 4 and 4 (c) 4 and 3**
**Why:** Work from the ones. (a) 4 + 9 = 13, so the bottom digit is 9 (carry 1). 4 + 5 + 1 = 10, so the top digit is 4. (b) 8 + 6 = 14 (carry 1). 4 + 2 + 1 = 7, so the top digit is 4. 4 + 4 = 8, so the bottom digit is 4. (c) 5 + 8 = 13 (carry 1). 6 + 3 + 1 = 10, so the bottom digit is 3. 4 + 2 + 1 = 7, so the top digit is 4.

**2** page 14
**Answer: (a) 3 and 7 (b) 1 and 2**
**Why:** Turn each into an addition. (a) 568 + 267 = 835: 15, 13, 8 with carries. (b) 265 + 247 = 512: 12, 11, 5 with carries. So 835 - 267 = 568 and 512 - 247 = 265.

**3** page 14
**Answer: (a) 110 (b) 740 (c) 190 (d) True**
**Why:** (a) 270 + 140 = 410, and 410 - 300 = 110. (b) 840 - 300 = 540, and 540 + 200 = 740. (c) 400 + 150 = 550, and 550 - 360 = 190. (d) 199 is 1 less than 200 and 250 is 1 more than 249, so both sides are 449.

**4** page 15
**Answer: (a) 750 (b) 350 (top), 450 (left), 50 (right), 150 (bottom)**
**Why:** A line with all 3 numbers: 100 + 250 + 400 = 750. Top row: 750 - 100 - 300 = 350. Left column: 750 - 100 - 200 = 450. Right column: 750 - 300 - 400 = 50. Bottom row: 750 - 200 - 400 = 150.

**5** page 15
**Answer: (a) 380 (b) 190 (c) 270**
**Why:** Without the extra 80, both have the same, and together they have 380. Half of 380 is 190, so Ben has 190. Ali has 190 + 80 = 270. Check: 190 + 270 = 460.

**6** page 15
**Answer: (a) 624 (b) No (c) 336**
**Why:** (a) 346 + 278 = 624. (b) A right answer would add back to 614, so Ben is wrong. (c) 614 - 278 = 336. Check: 336 + 278 = 614. Ben may have forgotten that the tens had lent 1 ten, so his tens digit is 1 too big.

**7** page 16
**Answer: (a) (2) (b) 217**
**Why:** (a) He starts with 345 and takes some away, so 345 - [ ] = 128, which is (2). (b) 345 - 128 = 217. Check: 128 + 217 = 345.

**8** page 16
**Answer: 155**
**Why:** Take away what she used and what was left: 500 - 135 - 210 = 155. Check: 155 + 135 + 210 = 500.

**9** page 16
**Answer: (a) 825 (b) 317 (c) 643 (d) 444**
**Why:** Add the ones, then the tens, then the hundreds, and carry. (a) 8 + 7 = 15, 6 + 5 + 1 = 12, 3 + 4 + 1 = 8: 825. (b) 702 - 385 = 317 (check 317 + 385 = 702). (c) 9 + 4 = 13, 4 + 9 + 1 = 14, 2 + 3 + 1 = 6: 643. (d) 910 - 466 = 444 (check 444 + 466 = 910).

**10** page 16
**Answer: (a) 660 (b) 515 (c) 245 (d) 925**
**Why:** (a) 1000 - 340 = 660. (b) 1000 - 485 = 515. (c) 1000 - 755 = 245. (d) 1000 - 75 = 925. Tip: make the ones 10, the tens 9 and the hundreds 9 to reach 999, then add 1.

**11** page 16
**Answer: (a) 213 (b) 93**
**Why:** (a) Count on: 287 + 13 = 300 and 300 + 200 = 500, so 13 + 200 = 213 pages are left. (b) 213 - 120 = 93 pages.

## THINK HARDER

**1** page 17
**Answer: (a) 120, 135, 150 (b) 165, 180, 195**
**Why:** 450: corners 120, 135, 150, and the sides are 195, 165 and 180. 495: the corners are the 3 greatest numbers 165, 180, 195, and the sides are 150, 120 and 135.

**2** page 17
**Answer: (a) True (b) False (c) True (d) False**
**Why:** (a) True: 1 more and 1 less. (b) False: 10 more but 20 less, so 590 and 600. (c) True: 2 more and 2 less. (d) False: 550 and 560.

**3** page 17
**Answer: (a) 170 (b) 230 (c) 130**
**Why:** Ali has 60 more and Siti 40 fewer, so the 3 have 20 more than 3 Bens. 530 - 20 = 510, and 510 ÷ 3 = 170 for Ben. Ali 230, Siti 130. Check: 230 + 170 + 130 = 530.

## GO FURTHER

**1** page 18
**Answer: (a) 190 (b) 350 (c) 260**
**Why:** 540 + 610 + 450 = 1600 counts each child twice, so all 3 have 800. Ravi 800 - 540 = 260. Ali 800 - 610 = 190. Mei 800 - 450 = 350.

**2** page 18
**Answer: (a) 495 and 693 (b) 9 (c) 495**
**Why:** (a) 742 - 247 = 495 and 831 - 138 = 693. (b) Both have 9 in the middle. (c) The answer is always a multiple of 99: 954 - 459 = 495 (5 × 99, and 9 - 4 = 5).

**3** page 18
**Answer: Open. Example: 4?6 + 37? = 801 (426 + 375). Test every digit for each box: only 2 and 5 work.**
**Why:** Example: 426 + 375 = 801. Ones: 6 + ? must end in 1, so the ones box is 5. That carries 1. Tens: ? + 7 + 1 must end in 0, so the tens box is 2. Check that no other digit fits in each box.

**4** page 18
**Answer: (a) 600 (b) 500**
**Why:** (a) Ali gets 900 - 102 = 798. Mei's number plus 198 is 798, so it is 798 - 198 = 600. (b) Siti gets 1000 - 201 = 799. Ravi's number is 799 - 299 = 500. Check: 500 + 299 = 799.

### CHECK YOURSELF

**1** page 19
**Answer: 5 and 3**
**Why:** Ones: 6 + 3 = 9, so the bottom box is 3. Tens: 5 + 3 = 8, so the top box is 5. Check: 256 + 433 = 689.

**2** page 19
**Answer: 90**
**Why:** 530 + 160 = 690, and 690 - 600 = 90.

**3** page 19
**Answer: 275**
**Why:** 800 - 175 = 625. 625 - 350 = 275. Check: 175 + 275 + 350 = 800.

**4** page 19
**Answer: 180**
**Why:** Take away the extra 60: 420 - 60 = 360. Half of 360 is 180. The numbers are 180 and 240.

## Unit 3 · Legs, wheels and arrays

**Your turn** page 20
**Answer: 2**
**Why:** Start with 3 ducks: 6 legs. Swap 1 duck for a dog: 8 legs. Swap again: 10 legs. So 1 duck and 2 dogs. Check: 2 + 8 = 10 legs.

### START HERE

**1** page 21
**Answer: (a) 3 × 7 = 21 and 7 × 3 = 21 (either order) (b) 21 ÷ 3 = 7 and 21 ÷ 7 = 3 (either order)**
**Why:** There are 3 rows with 7 counters in each row. 3 × 7 = 21 and 7 × 3 = 21. Share 21 into 3 rows: 21 ÷ 3 = 7. Put 21 in rows of 7: 21 ÷ 7 = 3.

**2** page 21
**Answer: (a) 8 (b) 32 (c) 8**
**Why:** (a) The top row has 8 cookies, and every row has the same number. (b) 4 rows of 8 is 32. (c) You can see 8 + 8 + 4 + 4 = 24 cookies. 32 - 24 = 8 are under the towel.

**3** page 21
**Answer: (a) True (b) False (c) True (d) False**
**Why:** (a) 4 plates with 5 counters each: true. (b) 5 + 5 + 5 is only 3 plates, so 15 is not all of them: false. (c) 4 groups of 5 is 4 × 5 = 20: true. (d) The picture shows 4 plates of 5, not 5 plates of 4: false.

**4** page 22
**Answer: (a) 12 (b) 4 (c) 16**
**Why:** (a) 3 groups of 4: 4 + 4 + 4 = 12. (b) 2 groups of 2: 2 + 2 = 4. (c) 12 + 4 = 16.

**5** page 22
**Answer: (a) 20, 22, 24, 26 (b) 3 ducks and 5 rabbits**
**Why:** Each swap adds 2 legs: 16, 18, 20, 22, 24, 26. 26 legs is 3 ducks and 5 rabbits. Check: 6 + 20 = 26 legs and 3 + 5 = 8 animals.

**6** page 22
**Answer: (a) 24 (b) 6 (c) 9 (d) 24**
**Why:** (a) 6 boxes of 4 pencils: 6 × 4 = 24. (b) Share 30 into 5 equal bags: 30 ÷ 5 = 6. (c) How many groups of 5 in 45? 45 ÷ 5 = 9. (d) 6 cars with 4 wheels each: 6 × 4 = 24.

**7** page 23
**Answer: (a) 7 (b) 2 (c) 3**
**Why:** (a) 7 × 5 = 35 and 8 × 5 = 40 is too many, so 7 boxes. (b) 37 - 35 = 2 left over. (c) 5 - 2 = 3 more biscuits fill another box.

**8** page 23
**Answer: (a) 5 (b) 3**
**Why:** (a) 4 minibuses hold 40 people. 7 people are left, and they need a minibus too, so 5 minibuses. (b) 5 minibuses hold 50 people. 50 - 47 = 3 empty seats.

**9** page 23
**Answer: (a) 9, 6, no, no, 3, no, no, 2 (b) 4**
**Why:** Rows that share 18 equally work: 2 rows of 9, 3 rows of 6, 6 rows of 3, 9 rows of 2. 4, 5, 7 and 8 rows leave some pots over, so no. 4 ways.

**10** page 23
**Answer: (a) 8 (b) 6 (c) 2 each and 4 left over (d) 6**
**Why:** (a) 24 ÷ 3 = 8. (b) 24 ÷ 4 = 6. (c) 10 × 2 = 20, so each gets 2 and 24 - 20 = 4 are left. (d) Siti and her 3 friends are 4 people, so 24 ÷ 4 = 6.

## THINK HARDER

**1** page 24
**Answer: (a) 3 (b) 8**
**Why:** Trikes have 3 wheels. 1 trike leaves 18 wheels: 9 scooters. 3 leave 12: 6 scooters. 5 leave 6: 3 scooters. 2, 4 and 6 leave an odd number and 7 leaves none. So 3 ways, with 10, 9 and 8 vehicles. The fewest is 8.

**2** page 24
**Answer: (a) Yes (b) No (c) No (d) Yes**
**Why:** All hens: 12 legs. Each swap adds 2, up to all pigs: 24. Totals: 12, 14 ... 24. (a) 18: yes. (b) 25 is odd: no. (c) 26 is over 24: no. (d) 12: yes.

**3** page 24
**Answer: 33**
**Why:** Groups of 3 with none left over: 30, 33, 36, 39, 42, 45, 48. Groups of 4 with 1 left: 33 and 45. Groups of 5 with 3 left: 33 has 3 left, 45 has none. So 33.

## GO FURTHER

**1** page 25
**Answer: (a) 36 (b) 48 (c) 9 (d) 42**
**Why:** (a) 30 + 6 = 36. (b) 6 × 8 is double 3 × 8 = 24, so 48. (c) 6 × 9 = 54, so 54 ÷ 6 = 9. (d) 36 + 6 = 42.

**2** page 25
**Answer: (a) 5 bags of 3 and 4 bags of 5 (b) No**
**Why:** (a) All 9 bags of 3 apples: 27 apples. Each swap of a 3-bag for a 5-bag adds 2: 29, 31, 33, 35 after 4 swaps. So 4 bags of 5 and 5 bags of 3. (b) The totals are 27, 29, 31 ... always odd. 40 is even, so it cannot be done.

**3** page 25
**Answer: Open. Example: 6 pieces and 21 legs gives 3 stools and 3 chairs (9 + 12 = 21). It is the only answer because each swap adds 1 leg.**
**Why:** Example: 3 stools and 3 chairs have 9 + 12 = 21 legs and 6 pieces. Starting with 6 stools (18 legs), each swap to a chair adds 1 leg, so 21 legs needs exactly 3 swaps.

**4** page 25
**Answer: 6**
**Why:** Rows that share 24 equally: 2 rows of 12, 3 rows of 8, 4 rows of 6, 6 rows of 4, 8 rows of 3, 12 rows of 2. 5, 7, 9, 10 and 11 rows do not work. That is 6 arrays.

### CHECK YOURSELF

**1** page 26
**Answer: 24**
**Why:** 3 rows of 8: 8 + 8 + 8 = 24.

**2** page 26
**Answer: 28**
**Why:** 5 dogs: 5 × 4 = 20 legs. 4 ducks: 4 × 2 = 8 legs. 20 + 8 = 28.

**3** page 26
**Answer: 4 boxes and 7 pencils**
**Why:** 4 boxes hold 40 pencils. 47 - 40 = 7 are left over.

**4** page 26
**Answer: 3**
**Why:** 9 birds have 18 legs. Each swap to a rabbit adds 2 legs: 20, 22, 24 after 3 swaps. So 3 rabbits and 6 birds. Check: 12 + 12 = 24.

## Unit 4 · Pattern machines

**Your turn** page 27
**Answer: square**
**Why:** One group has 3 shapes. 12 is in the 3 times table, so place 12 is the last shape of a group. Place 13 is a star and place 14 is a square.

### START HERE

**1** page 28
**Answer: 45, 40, 35**
**Why:** Take away 5 each time: 50 - 5 = 45, 45 - 5 = 40, 40 - 5 = 35. These numbers are in the 5 times table.

**2** page 28
**Answer: (a) circle (b) square (c) star**
**Why:** One group has 4 shapes. (a) 16 ends a group: a circle. (b) 20 ends a group, so 21 is a star and 22 is a square. (c) 32 ends a group, so 33 is a star.

**3** page 28
**Answer: (a) H (b) 28 (c) 12**
**Why:** One group is C, H, E, S, S. (a) 15 ends a group, so 16 is C and 17 is H. (b) E is 3rd in a group: 3, 8, 13, 18, 23, 28. (c) 6 groups with 2 S each: 12.

**4** page 28

**Answer: (a) 32 (b) 40**
**Why:** One group has 1 square and 4 round beads, 5 beads in all. 8 square beads means 8 groups. (a) 8 groups of 4 round beads: 8 × 4 = 32. (b) 8 groups of 5 beads: 8 × 5 = 40. Check: 8 + 32 = 40.

**5** page 29

**Answer: (a) 8, 12, 16, 20, 24 (b) 40 (c) 7 (d) 3 pizzas and 2 slices**
**Why:** Each pizza adds 4 slices: 4, 8, 12, 16, 20, 24. (b) 10 × 4 = 40. (c) 4 × 7 = 28. (d) 3 pizzas use 12 slices, and 14 - 12 = 2 slices are left over.

**6** page 29

**Answer: (a) 13 (b) 19 (c) 25**
**Why:** Each new square adds 3 sticks: 4, 7, 10, 13, 16, 19, 22, 25. (a) 13. (b) 19. (c) 25. Another way: 8 squares need 8 × 3 = 24 sticks, plus 1 extra for the first side. 24 + 1 = 25.

**7** page 29

**Answer: (a) 15 and 30 (b) 45**
**Why:** Siti: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39. Ravi: 5, 10, 15, 20, 25, 30, 35, 40. Both say 15 and 30. They meet every 15, so the next is 30 + 15 = 45.

**8** page 30

**Answer: (a) 27, 32 and add 5 (b) 32, 26 and subtract 6**
**Why:** (a) Each number is 5 more: 22 + 5 = 27 and 27 + 5 = 32. (b) Each number is 6 less: 38 - 6 = 32 and 32 - 6 = 26.

**9** page 30

**Answer: (a) 8 (b) 16 (c) 8**
**Why:** (a) 24 ÷ 3 = 8 groups. (b) Each group has 2 circles: 8 × 2 = 16. (c) Each group has 1 triangle: 8 × 1 = 8. Check: 16 + 8 = 24.

**10** page 30

**Answer: (a) 11, 13, 15 (b) 23 (c) 8**
**Why:** Each tower has 2 more cubes: 5, 7, 9, 11, 13, 15, 17, 19, 21, 23. Tower 10 has 23 cubes and tower 8 has 19 cubes. Another way: tower n has n × 2 + 3 cubes, so tower 10 has 20 + 3 = 23.

## THINK HARDER

**1** page 31

**Answer: (a) Yes (b) No (c) Yes**
**Why:** Every 2 hops the frog goes 4 further: 3, 4, 7, 8, 11, 12, 15, 16 ... It lands on the 4 times table and on the number 3 before each one. 27 = 24 + 3: yes. 26 is between 24 and 27: no. 40 = 4 × 10: yes.

**2** page 31

**Answer: (a) Yes (b) Yes (c) No (d) Yes**
**Why:** The star is the 2nd of 5 shapes: places 2, 7, 12, 17, 22 ... (2 more than the 5 times table). 17 and 22: yes. 35 is the last shape: no. 47 = 45 + 2: yes.

**3** page 31

**Answer: (a) 22 (b) 42 (c) 7**
**Why:** Each carriage adds 5 children, and the engine holds 2: 2, 7, 12, 17, 22 ... (a) 4 carriages: 2 + 20 = 22. (b) 8 carriages: 2 + 40 = 42. (c) 37 - 2 = 35 children are in carriages, and 35 ÷ 5 = 7 carriages.

### GO FURTHER

**1** page 32
**Answer: (a) 36 (b) 36 and 49 (c) 100**
**Why:** (a) 6 × 6 = 36. (b) After 5 × 5 = 25 come 6 × 6 = 36 and 7 × 7 = 49. (c) 10 × 10 = 100. The gaps grow by 2 each time: 3, 5, 7, 9 ...

**2** page 32
**Answer: (a) 4, 16 and 28 (b) 40 (c) No**
**Why:** Frog A: 4, 8, 12 ... 40. Frog B: 1, 4, 7 ... 40. Both: 4, 16, 28, then 40: every 12, because 12 is in the 3 and 4 times tables. Frog C lands 2 more than the 4 times table, so it never meets Frog A.

**3** page 32
**Answer: (a) Siti (b) circle**
**Why:** One group has 5 shapes and 40 is in the 5 times table, so place 40 is the last shape of a group: a square. Siti is right. Place 41 is a star, place 42 and place 43 are circles.

**4** page 32
**Answer: Open. Example: triangle, square, circle, circle. Place 28 ends a group, so place 29 is a triangle and place 30 is a square.**
**Why:** Example: triangle, square, circle, circle repeats. Groups end at 4, 8, ... 28. Place 28 is a circle, so place 29 is a triangle and place 30 is a square.

### CHECK YOURSELF

**1** page 33
**Answer: circle**
**Why:** One group has 3 shapes. 18 is in the 3 times table, so place 18 is a square. Place 19 is a star and place 20 is a circle.

**2** page 33
**Answer: 45**
**Why:** 9 groups of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45.

**3** page 33
**Answer: 2**
**Why:** Ali says 4, 8, ... 40. Mei says 5, 10, ... 40. Both say 20 and 40. That is 2 numbers.

**4** page 33
**Answer: 23**
**Why:** Each tower has 4 more: 3, 7, 11, 15, 19, 23. Tower 6 has 23 cubes.

## Unit 5 · Mystery number machines

**Your turn** page 34
**Answer: 8**
**Why:** Undo - 8 with + 8: 32 + 8 = 40. Undo × 5 with ÷ 5: 40 ÷ 5 = 8. Check: 8 × 5 = 40 and 40 - 8 = 32.

### START HERE

**1** page 35
**Answer: (a) 34 (b) 7 (c) 36 (d) 43**
**Why:** Use the opposite step. (a) 61 - 27 = 34. (b) 35 ÷ 5 = 7. (c) 9 × 4 = 36. (d) 25 + 18 = 43.

**2** page 35
**Answer: (a) - 7 (b) ÷ 5 (c) + 12 (d) × 3**
**Why:** The opposite of adding is taking away, and the opposite of multiplying is dividing. (a) - 7. (b) ÷ 5. (c) + 12. (d) × 3.

**3** page 35
**Answer: (a) 8 (b) 13 (c) 20**
**Why:** (a) 25 - 9 = 16, then 16 ÷ 2 = 8. Check: 8 × 2 = 16, 16 + 9 = 25. (b) 28 ÷ 4 = 7, then 7 + 6 = 13. Check: 13 - 6 = 7, 7 × 4 = 28. (c) 11 - 7 = 4, then 4 × 5 = 20. Check: 20 ÷ 5 = 4, 4 + 7 = 11.

**4** page 35
**Answer: (a) 16 (b) 6 (c) 7**
**Why:** Undo the last step first. (a) 15 - 7 = 8, then 8 × 2 = 16. (b) 19 + 5 = 24, then 24 ÷ 4 = 6. (c) 78 - 8 = 70, then 70 ÷ 10 = 7.

**5** page 36
**Answer: (a) 11 and 23 in the OUT row, 8 and 10 in the IN row**
**Why:** IN 3: 3 × 4 = 12, 12 - 1 = 11. IN 6: 6 × 4 = 24, 24 - 1 = 23. OUT 31: 31 + 1 = 32, 32 ÷ 4 = 8. OUT 39: 39 + 1 = 40, 40 ÷ 4 = 10.

**6** page 36
**Answer: (a) 2 (b) 5 (c) 25**
**Why:** Try 2 stickers a token: 3 tokens give 6, and 11 - 6 = 5 is the bonus. Test 6 tokens: 12 + 5 = 17. It fits. 3 and 4 a token do not fit both. 10 tokens: 20 + 5 = 25.

**7** page 36
**Answer: (a) 35 (b) 53 (c) 35, then 60**
**Why:** Undo the last step: 60 - 25 = 35. Undo the first step: 35 + 18 = 53. Check: 53 - 18 = 35, then 35 + 25 = 60.

**8** page 37
**Answer: (a) 10 (b) 40**
**Why:** Box 4 had 7 + 3 = 10 stickers before he took 3 out. Every box had the same, so each box had 10. 4 boxes: 4 × 10 = 40 stickers.

**9** page 37
**Answer: (a) 20 (b) 8**
**Why:** 4 albums with 5 stickers each: 4 × 5 = 20 stickers. That is after buying 12 more, so at first she had 20 - 12 = 8. Check: 8 + 12 = 20 and 20 ÷ 4 = 5.

**10** page 37
**Answer: (a) 11 (b) 23**
**Why:** (a) $3 + $8 = $11 spent. (b) Undo the spending: $12 + $11 = $23. Check: 23 - 3 = 20, 20 - 8 = 12.

## THINK HARDER

**1** page 38
**Answer: (a) 12 (b) 18 (c) 21**
**Why:** Take away the extra 6 and 9: 51 - 6 - 9 = 36. Now 3 equal bars share 36, so each is 36 ÷ 3 = 12. Mei has 12, Lily 12 + 6 = 18 and Aisha 12 + 9 = 21. Check: 12 + 18 + 21 = 51.

**2** page 38
**Answer: (a) 15 (b) 23**
**Why:** Now she has twice the number she had after the card, so 30 is 2 equal parts: 30 ÷ 2 = 15. After the card she had 15. At first she had 15 + 8 = 23. Check: 23 - 8 = 15, then 15 + 15 = 30.

**3** page 38
**Answer: (a) Yes (b) No (c) Yes (d) No**
**Why:** IN 1, 2, 3, 4 give 5, 9, 13, 17: always 1 more than the 4 times table. 29 = 28 + 1: yes. 30 is skipped (29, 33): no. 33 = 32 + 1: yes. 42 is skipped (41, 45): no.

### GO FURTHER

**1** page 39
**Answer: (a) 7 (b) 6 (c) 10**
**Why:** (a) 25 - 4 = 21, 21 ÷ 3 = 7. (b) 27 + 3 = 30, 30 ÷ 5 = 6. (c) 13 - 8 = 5, 5 × 2 = 10.

**2** page 39
**Answer: (a) 12 (b) 16, 14 and 6**
**Why:** (a) 36 ÷ 3 = 12. (b) Undo B to C: jar C had 12 - 6 = 6 and jar B had 12 + 6 = 18. Undo A to B: jar B had 18 - 4 = 14 and jar A had 12 + 4 = 16. Check: 16 + 14 + 6 = 36.

**3** page 39
**Answer: Open. Example: I think of a number. I add 7, then multiply by 2. I get 38. (38 ÷ 2 = 19, 19 - 7 = 12.)**
**Why:** Start at 12 and go forwards: 12 + 7 = 19, 19 × 2 = 38. The riddle ends at 38. To solve it, work backwards from 38.

**4** page 39
**Answer: (a) 17 (b) 7**
**Why:** Undo B: 36 ÷ 3 = 12, then 12 + 5 = 17. That is the OUT of A. Undo A: 17 - 3 = 14, then 14 ÷ 2 = 7. Check: 7 → 14 → 17 → 12 → 36.

### CHECK YOURSELF

**1** page 40
**Answer: 8**
**Why:** 38 - 6 = 32, then 32 ÷ 4 = 8. Check: 8 × 4 = 32, 32 + 6 = 38.

**2** page 40
**Answer: 36**
**Why:** Undo buying 9: 31 - 9 = 22. Undo giving away 14: 22 + 14 = 36. Check: 36 - 14 = 22, 22 + 9 = 31.

**3** page 40
**Answer: 8**
**Why:** 25 - 9 = 16, then 16 ÷ 2 = 8. Check: 8 × 2 = 16, 16 + 9 = 25.

**4** page 40
**Answer: 50**
**Why:** Bag 5 had 4 + 6 = 10 pens before. Every bag had 10, so 5 × 10 = 50 pens.

## Unit 6 · Market day

**Your turn** page 41
**Answer: 6**
**Why:** $6 is $5 + $1 (2 pieces). 85¢ is 50¢ + 20¢ + 10¢ + 5¢ (4 pieces). 2 + 4 = 6 pieces. Check: 5 + 1 + 0.50 + 0.20 + 0.10 + 0.05 = 6.85.

### START HERE

**1** page 42
**Answer: (a) 345 (b) 2.75 (c) 6.05 (d) 130**
**Why:** $1 is 100¢. (a) 3 dollars is 300¢, and 300 + 45 = 345. (b) 275¢ is 2 dollars and 75 cents: $2.75. (c) 6 dollars and 5 cents is $6.05, not $6.50. (d) 100¢ + 30¢ = 130¢.

**2** page 42
**Answer: (a) 10.80**
**Why:** Notes and $1: $5 + $2 + $2 + $1 = $10. Coins: 50¢ + 20¢ + 10¢ = 80¢. $10 + $0.80 = $10.80.

**3** page 42
**Answer: (a) 5 (b) 4 (c) 7**
**Why:** Marker pen: $2 + 50¢ + 20¢ + 10¢ + 5¢ is 5 pieces. Kite: $5 + $2 + 20¢ + 20¢ is 4 pieces. Backpack: $10 + $10 + $2 + $1 + 50¢ + 10¢ + 5¢ is 7 pieces (there is no $20 note).

**4** page 43
**Answer: (a) 6 (b) 3 $2 notes and 1 $1 coin**
**Why:** Ways ($5, $2, $1): 1, 1, 0 / 1, 0, 2 / 0, 3, 1 / 0, 2, 3 / 0, 1, 5 / 0, 0, 7. That is 6 ways. Their pieces are 2, 3, 4, 5, 6 and 7. Only 0, 3, 1 has 4 pieces: three $2 notes and one $1 coin.

**5** page 43
**Answer: (a) 30 (b) 5.70 (c) 13 (d) 3**
**Why:** Change is what you pay minus the cost. (a) 100¢ - 70¢ = 30¢. (b) $10 - $4.30 = $5.70. Count on: $4.30 to $5 is 70¢, then $5 more makes $10. (c) $50 - $37 = $13. (d) $13 is $10 + $2 + $1: 3 pieces.

**6** page 44
**Answer: (a) 27.40 (b) No (c) 2.00**
**Why:** (a) $17.50 + $9.90 = $27.40. (b) All 3 cost $27.40 + $4.60 = $32.00. Ravi has only $30, so no. (c) $32.00 - $30 = $2.00 more.

**7** page 44
**Answer: 9.50**
**Why:** Work backwards: put back what she spent. $4.75 + $1.25 = $6.00, and $6.00 + $3.50 = $9.50. Check: 9.50 - 3.50 = 6.00 and 6.00 - 1.25 = 4.75.

**8** page 44
**Answer: (a) 3 (b) 8 (c) 15**
**Why:** (a) 12 pencils is 3 groups of 4, so 3 × $1 = $3. (b) 20 stickers is 4 groups of 5, so 4 × $2 = $8. (c) $6 is 3 groups of $2, so 3 × 5 = 15 stickers.

**9** page 44
**Answer: Car and Top**
**Why:** $4.50 + $8.00 = $12.50. Check the others: Car + Ball = $10.75, Car + Kite = $8.25, Ball + Top = $14.25, Ball + Kite = $10.00, Top + Kite = $11.75. Only Car and Top.

**10** page 44
**Answer: 50¢ + 20¢ + 20¢ + 5¢ (any order)**
**Why:** Start with 50¢. 45¢ is left for 3 coins. 20¢ + 20¢ = 40¢, and 5¢ makes 45¢. Check: 50 + 20 + 20 + 5 = 95. No other 4 coins make 95¢.

## THINK HARDER

**1** page 45
**Answer: (a) 6 (b) 2 coins of 10¢ and 3 coins of 5¢**
**Why:** With one 20¢: 15¢ is 10+5 or 5+5+5. With no 20¢: 35¢ is made of 10¢ and 5¢ coins. Ways (20, 10, 5): 1,1,1 / 1,0,3 / 0,3,1 / 0,2,3 / 0,1,5 / 0,0,7: 6 ways. Only 0, 2, 3 has 5 coins.

**2** page 45
**Answer: (a) 3 (b) 4**
**Why:** 7 lollipops cost $1.40. Each swap of a lollipop for a cookie adds 30¢: $1.70, $2.00, $2.30, $2.60. That is 4 swaps. So 3 lollipops and 4 cookies. Check: 60¢ + $2.00 = $2.60 and 3 + 4 = 7.

**3** page 45
**Answer: (a) 4.50 (b) 8.50**
**Why:** Take away the extra $4: $13 - $4 = $9. Two equal bars share $9, so each is $4.50. The puzzle costs $4.50 and the book costs $4.50 + $4 = $8.50. Check: 4.50 + 8.50 = 13.00.

### GO FURTHER

**1** page 46
**Answer: (a) 4.80 (b) 7.25 (c) 11.60**
**Why:** (a) Half of $9.60 is $4.80. (b) Half of $14 is $7 and half of 50¢ is 25¢: $7.25. (c) The tag is double: $5.80 + $5.80 = $11.60.

**2** page 46
**Answer: (a) 6 (b) 1.90 (c) No**
**Why:** 5 coins of 20¢ is $1.00. Each swap of a 20¢ for a 50¢ adds 30¢: $1.00, $1.30, $1.60, $1.90, $2.20, $2.50. That is 6 amounts. (b) 3 coins of 50¢ and 2 of 20¢: $1.50 + $0.40 = $1.90. (c) $1.40 is not in the list: no.

**3** page 46
**Answer: Open. Example: 3 coins are worth 80¢. Only 50¢, 20¢ and 10¢ fit.**
**Why:** Example: 3 coins worth 80¢. The only set is 50¢ + 20¢ + 10¢. Any other 3 coins make a different total. A riddle is good if no other set fits.

**4** page 46
**Answer: (a) 8 (b) 9 (c) A**
**Why:** (a) 12 pens is 4 groups of 3, so 4 × $2 = $8. (b) 12 pens is 3 groups of 4, so 3 × $3 = $9. (c) $8 is less than $9, so shop A is cheaper by $1.

### CHECK YOURSELF

**1** page 47
**Answer: 405**
**Why:** 4 dollars is 400¢. 400¢ + 5¢ = 405¢.

**2** page 47
**Answer: 5**
**Why:** $6 is $5 + $1 (2 pieces). 35¢ is 20¢ + 10¢ + 5¢ (3 pieces). 2 + 3 = 5 pieces.

**3** page 47
**Answer: 2.60**
**Why:** $10 - $7.40 = $2.60. Count on: $7.40 + 60¢ = $8, then $2 more makes $10.

**4** page 47
**Answer: 10.00**
**Why:** Put back what he spent: $4.50 + $3.10 = $7.60, and $7.60 + $2.40 = $10.00.

## Unit 7 · Fair shares

**Your turn** page 48
**Answer: 7/10**
**Why:** 10 - 3 = 7 slices are left. Each slice is 1/10, so 7/10 is left. Check: 3/10 + 7/10 = 10/10 = 1 whole.

### START HERE

**1** page 49
**Answer: (a) Yes (b) No (c) Yes (d) No (e) Yes (f) No**
**Why:** Quarters must be 4 parts of the SAME size. A, C and E are quarters. In B and F one part is much greater than the other three. In D the parts are not the same size. In E the four triangles are all the same size.

**2** page 49
**Answer: (a)** 2/5 and 3/5 **(b)** 6/8 and 2/8 **(c)** 5/12 and 7/12 **(d)** 3
**Why:** The bottom number is the number of equal parts. (a) 2/5 and 3/5. (b) 6/8 and 2/8. (c) 5/12 and 7/12. (d) 8/12 needs 8 shaded parts. 5 are shaded, so 3 more.

**3** page 49
**Answer: (a)** 1 **(b)** 9 **(c)** 4
**Why:** (a) 6 sixths make 1 whole. (b) 1 whole cut into 9 equal parts is 9/9. (c) 1 whole is 8/8, and 8 - 4 = 4, so 4/8 + 4/8 = 1.

**4** page 50
**Answer: (a)** 1/3 **(b)** 1/8 **(c)** 1/8, 1/6, 1/4, 1/2
**Why:** The more parts a bar is cut into, the smaller each part. (a) Thirds are greater than quarters. (b) Eighths are smaller than sixths. (c) Order: 1/8, 1/6, 1/4, 1/2.

**5** page 50
**Answer: (a)** 6/10 **(b)** 7/12 **(c)** 2/7, 4/7, 6/7
**Why:** When the bottom numbers are the same, the parts are the same size, so more parts is greater. (a) 6/10 is greater than 3/10. (b) 7/12 is greater than 5/12. (c) 2/7, 4/7, 6/7.

**6** page 50
**Answer: (a)** 9/12 **(b)** 5/8 **(c)** 2/5 **(d)** 5/9
**Why:** (a) 3 twelfths + 6 twelfths = 9 twelfths. (b) 7 eighths - 2 eighths = 5 eighths. (c) 1 whole is 5/5, and 5/5 - 3/5 = 2/5. (d) 1 whole is 9/9, and 9/9 - 4/9 = 5/9.

**7** page 51
**Answer: (a)** 7/10 **(b)** 3/10 **(c)** Ravi **(d)** 1/10
**Why:** Siti ate 3 slices and Ravi ate 4 slices. (a) 3/10 + 4/10 = 7/10. (b) 10/10 - 7/10 = 3/10. (c) 4 is more than 3, so Ravi. (d) 4/10 - 3/10 = 1/10.

**8** page 51
**Answer: (a)** 1/5 **(b)** 2/5 **(c)** 3/5
**Why:** (a) 5 equal shares: each is 1/5. (b) 1/5 + 1/5 = 2/5. (c) The whole is 5/5. 5/5 - 2/5 = 3/5.

**9** page 51
**Answer: (a)** 3/8 **(b)** 7/8
**Why:** Each step is 1/8. (a) A is 3 steps from 0: 3/8. (b) B is 7 steps from 0: 7/8.

## THINK HARDER

**1** page 52
**Answer: (a)** Ravi **(b)** 1/8
**Why:** The bars are the same length. 3/4 of the first bar covers 6 of the small eighths, because each quarter is 2 eighths. Siti eats 5 eighths. Ravi eats 6 eighths, so Ravi eats 1/8 more.

**2** page 52
**Answer: (a)** 3/4 **(b)** 3/8 **(c)** 3/8
**Why:** (a) 1/4 eaten, so 3/4 is left. On the bar of 8 parts, 1/4 is 2 parts, and 6 parts are left. (b) Half of 6 parts is 3 parts: Ali eats 3/8. (c) 6 - 3 = 3 parts are left: 3/8.

**3** page 52
**Answer: (a)** 3/12 **(b)** 9/12
**Why:** The whole is 12/12. Take away the extra 6/12: 6/12 is left. Share it between the two equal parts: 3/12 and 3/12. Give the extra back to one of them: 3/12 and 9/12. Check: 3/12 + 9/12 = 12/12.

### GO FURTHER

**1** page 53
**Answer: (a) 3 (b) 4 (c) 6**
**Why:** (a) A half of 6 sixths is 3 sixths. (b) A third is 2 sixths, so 2 thirds are 4 sixths. (c) A quarter is 2 eighths, so 3 quarters are 6 eighths.

**2** page 53
**Answer: (a) 3 (b) 1/9 and 2/9**
**Why:** The 3 different numbers of pieces add to 9. With 1 piece for the smallest: 1 + 2 + 6 and 1 + 3 + 5. With 2 pieces: 2 + 3 + 4. 1 + 4 + 4 repeats a number. So 3 ways. The way with 6/9 is 1/9, 2/9 and 6/9.

**3** page 53
**Answer: Open. Example: Ali eats 3/8, Mei eats 2/8, Ben eats 1/8 and 2/8 is left. 3 + 2 + 1 + 2 = 8 eighths = 1 whole.**
**Why:** A good story has shares and leftovers that add to 8/8. Example: 3/8 + 2/8 + 1/8 + 2/8 = 8/8 = 1 whole.

**4** page 53
**Answer: (a) 1/4, 1/5, 1/6, 1/7, 1/8 (b) 5**
**Why:** With 1 on top, the greater the bottom number, the smaller the fraction. 1/3 is too big and 1/9 is too small. 1/4, 1/5, 1/6, 1/7 and 1/8 are in between: 5 answers.

### CHECK YOURSELF

**1** page 54
**Answer: 5/8**
**Why:** The bar has 8 equal parts. 5 are shaded. The fraction is 5/8.

**2** page 54
**Answer: 1/5**
**Why:** The more parts a whole is cut into, the smaller each part. 5 parts are greater than 9 parts, so 1/5 is greater.

**3** page 54
**Answer: 6/7**
**Why:** 2 sevenths + 4 sevenths = 6 sevenths. The bottom number stays 7.

**4** page 54
**Answer: 4/12**
**Why:** 5/12 + 3/12 = 8/12 eaten. 12/12 - 8/12 = 4/12 is left.

## Unit 8 · Shape builders

**Your turn** page 55
**Answer: 8**
**Why:** Small squares: 2 rows of 3 is 6. Squares 2 by 2: 2 of them. 6 + 2 = 8 squares. There is no 3 by 3 square because the grid is only 2 high.

### START HERE

**1** page 56
**Answer: (a) 6 (b) 2 (c) 1 (d) 0**
**Why:** A cube has 6 flat square faces. A cylinder has 2 flat faces (circles) and 1 curved face. A cone has 1 flat face (a circle) and 1 curved face. A sphere has no flat face: it is all curved.

**2** page 56
**Answer: (a) 3 (b) 4 (c) 2 (d) 3**
**Why:** Go through the pieces one kind at a time. Half circles: 3. Quarter circles: 4. Squares: 2. Triangles: 3. Check: 3 + 4 + 2 + 3 = 12 pieces.

**3** page 56

**Answer: (a) 2 (b) 6 (c) 3**
**Why:** (a) 2 quarters make a half. (b) Each half is 2 quarters, so 3 halves are 6 quarters. (c) 4 quarters make 1 circle, so 12 quarters make 3 circles.

**4** page 57

**Answer: (a) 9 (b) 4 (c) 1 (d) 14**
**Why:** 1 by 1 squares: 3 rows of 3 is 9. 2 by 2 squares: they can start in 2 places across and 2 places down, so 4. 3 by 3 square: the whole grid, 1. Total: 9 + 4 + 1 = 14.

**5** page 57

**Answer: (a) 6 (b) 6 (c) 10**
**Why:** A: 3 + 2 + 1 = 6. B has 2 rows, and each row has 2 + 1 = 3 cubes, so 2 × 3 = 6. C has 2 rows of 2 + 2 + 1 = 5 cubes, so 2 × 5 = 10.

**6** page 57

**Answer: (a) 3 (b) 2 (c) 1 (d) 6**
**Why:** The 2 lines from the top cut the triangle into 3 small triangles. 2 small ones next to each other make a larger triangle: there are 2 such pairs. The whole triangle is 1 more. 3 + 2 + 1 = 6.

**7** page 58

**Answer: (a) cube (b) cone (c) sphere (d) cylinder**
**Why:** (a) Only a cube has 6 square faces. (b) A cone has one flat circle (its base) and one curved face. (c) A sphere is curved all over. (d) A cylinder has 2 flat circles, one at each end.

**8** page 58

**Answer: (a) circle (b) rectangle (c) circle**
**Why:** (a) 2 half circles make 1 whole circle. (b) 2 same squares joined side by side make a longer shape with 4 square corners: a rectangle. (c) 4 quarter circles make a whole circle.

**9** page 58

**Answer: (a) 10 (b) 30**
**Why:** (a) A layer is 2 rows of 5: 5 + 5 = 10 cubes. (b) 3 layers of 10 cubes: 10 + 10 + 10 = 30 cubes.

**10** page 58

**Answer: (a) square (b) circle (c) circle**
**Why:** (a) A cube has 6 faces and each one is a square. (b) The 2 flat faces of a cylinder are circles. (c) The flat face of a cone is its base, a circle.

## THINK HARDER

**1** page 59

**Answer: (a) Yes (b) Yes (c) No (d) No**
**Why:** A is a cross and B is a staircase of 3 pairs: both fold into a cube. C has a row of 5 squares: the 5th lands on the 1st face. D has a 2 by 2 block: 4 squares cannot all be different faces. So C and D do not fold.

**2** page 59

**Answer: (a) 5 (b) 4 (c) 3 (d) 3 and 15 in all**
**Why:** 1 stamp: 5 pieces. 2 stamps: they start at stamp 1, 2, 3 or 4, so 4. 3 stamps: 3. 4 stamps: 2. 5 stamps: 1. So 4 or 5 stamps: 2 + 1 = 3. All the pieces: 5 + 4 + 3 + 2 + 1 = 15.

**3** page 59

**Answer: (a) 6 (b) 6 (c) 12**
**Why:** The 2 lines from the top and the 2 sloping sides make 4 lines. Any 2 of them make a triangle on each line across: 6. There are 2 lines across, the base and the middle: 6 + 6 = 12 triangles.

### GO FURTHER

**1** page 60
**Answer: (a) 6 (b) 21 (c) 36**
**Why:** (a) Row 6 has 6 cans. (b) 1 + 2 + 3 + 4 + 5 + 6 = 21. (c) 21 + 7 + 8 = 36 (the rows 7 and 8 have 7 and 8 cans).

**2** page 60
**Answer: (a) 8 (b) 24 (c) 24 (d) 8**
**Why:** 3 faces: the 8 corners. 2 faces: 12 edges with 2 cubes each: 24. 1 face: 6 faces with a 2 by 2 middle: 24. No paint: the 2 by 2 by 2 middle: 8. Check: 8 + 24 + 24 + 8 = 64.

**3** page 60
**Answer: Open. Examples: 4 long, 3 wide and 2 high is 12 + 12 = 24. Or layers of 12, 8 and 4 cubes: 12 + 8 + 4 = 24.**
**Why:** Example 1: 4 by 3 by 2: each layer has 12 cubes and there are 2 layers: 24. Example 2: a layer of 4 by 3 (12), a layer of 4 by 2 (8) and a layer of 2 by 2 (4): 12 + 8 + 4 = 24.

**4** page 60
**Answer: 18**
**Why:** 1 by 1: 6. 1 high and 2 long: 4. 1 high and 3 long: 2. 2 high and 1 long: 3. 2 high and 2 long: 2. 2 high and 3 long: 1. Total: 6 + 4 + 2 + 3 + 2 + 1 = 18.

### CHECK YOURSELF

**1** page 61
**Answer: cone**
**Why:** A cone has a flat circle for its base and one curved face. A cylinder has 2 flat faces.

**2** page 61
**Answer: 6**
**Why:** 1 half circle is 2 quarter circles. 3 half circles make 3 × 2 = 6 quarter circles.

**3** page 61
**Answer: 24**
**Why:** One layer is 4 × 3 = 12 cubes. There are 2 layers: 12 + 12 = 24.

**4** page 61
**Answer: 8**
**Why:** Small squares: 6. Squares 2 by 2: 2. Total: 6 + 2 = 8.

## Unit 9 · Measure masters

**Your turn** page 62
**Answer: 100**
**Why:** The tin is as heavy as 4 pens. 4 × 25 g = 100 g. Check: 25 + 25 + 25 + 25 = 100.

### START HERE

**1** page 63
**Answer: (a) 4, 35 (b) 8, 50 (c) 11, 15**
**Why:** The long hand shows the minutes: count in 5s. A: long hand on 7 is 35 minutes, short hand past 4: 4:35. B: long hand on 10 is 50 minutes: 8:50. C: long hand on 3 is 15 minutes: 11:15.

**2** page 63
**Answer: (a) am (b) pm (c) pm (d) am**
**Why:** am is from midnight to noon. pm is from noon to midnight. (a) Breakfast is in the morning: am. (b) 1:30 after lunch: pm. (c) Bedtime at night: pm. (d) Sunrise is in the morning: am.

**3** page 63
**Answer: (a) 4, 15 (b) 40 (c) 4, 40**
**Why:** (a) 3:40 to 4:00 is 20 minutes. 15 more minutes: 4:15. (b) 9:50 to 10:00 is 10 minutes, and 10:00 to 10:30 is 30 minutes: 40 minutes. (c) Count back 5 minutes to 5:00, then 20 more minutes: 4:40.

**4** page 64
**Answer: (a) 1 hour 30 minutes**
**Why:** 9:35 to 10:00 is 25 minutes. 10:00 to 11:00 is 1 hour. 11:00 to 11:05 is 5 minutes. 25 + 5 = 30 minutes. The trip is 1 hour 30 minutes.

**5** page 64
**Answer: (a) 50 (b) 150**
**Why:** Scale 2: 1 lemon is 50 g. Scale 1: 1 apple is as heavy as 3 lemons: 3 × 50 = 150 g.

**6** page 64
**Answer: (a) 3 (b) 2 (c) 6**
**Why:** (a) 8 - 5 = 3 ℓ is left in the 8 ℓ jug. (b) The 5 ℓ jug holds 3 ℓ. 5 - 3 = 2 ℓ more fill it. (c) The 8 ℓ jug gives 2 ℓ: 8 - 2 = 6 ℓ is left.

**7** page 64
**Answer: (a) g (b) cm (c) kg (d) ℓ**
**Why:** An egg is light: grams. A pencil is short: centimetres. A school bag is heavy: kilograms. Water in a jug is measured in litres.

**8** page 65
**Answer: (a) 58 (b) 139 (c) 25**
**Why:** (a) 85 - 27 = 58 cm. (b) Together 85 + 58 = 143 cm. The overlap is counted twice, so take away 4 cm: 139 cm. (c) 3 × 20 = 60 cm cut off. 85 - 60 = 25 cm left.

**9** page 65
**Answer: (a) 3000 (b) 2, 500 (c) 1250**
**Why:** 1 kg is 1000 g. (a) 3 × 1000 = 3000 g. (b) 2500 g is 2000 g and 500 g: 2 kg 500 g. (c) 2 kg is 2000 g. 2000 - 750 = 1250 g.

**10** page 65
**Answer: (a) 235 (b) 1, 80 (c) 370**
**Why:** 1 m is 100 cm. (a) 200 + 35 = 235 cm. (b) 180 cm is 100 cm and 80 cm: 1 m 80 cm. (c) 5 m is 500 cm. 500 - 130 = 370 cm.

**11** page 65
**Answer: (a) 4 (b) 10**
**Why:** (a) The bucket needs 12 - 4 = 8 ℓ more. 8 ÷ 2 = 4 pours. (b) 3 pours add 6 ℓ. 4 + 6 = 10 ℓ.

## THINK HARDER

**1** page 66
**Answer: (a) 100 (b) 400**
**Why:** Swap the pear for 4 plums. Then 4 plums and 2 plums, which is 6 plums, balance 600 g. 600 ÷ 6 = 100, so 1 plum is 100 g. The pear is 4 plums: 400 g. Check: 400 + 100 + 100 = 600.

**2** page 66
**Answer: (a) 2, 7 (b) 4**
**Why:** Step 4: pour the 2 ℓ into the 5 ℓ jug (2, 0). Step 5: fill the 7 ℓ jug (2, 7). Step 6: pour from the 7 ℓ jug into the 5 ℓ jug until it is full. It needs 3 ℓ more, so 7 - 3 = 4 ℓ is left (5, 4). The 7 ℓ jug holds 4 ℓ.

**3** page 66
**Answer: (a) 95 (b) 10, 50**
**Why:** 4 stops of 20 minutes is 80 minutes. There are only 3 walks between 4 stops: 3 × 5 = 15 minutes. 80 + 15 = 95 minutes. 9:15 plus 95 minutes: 9:15 to 10:15 is 60 minutes, then 35 more: 10:50 am.

### GO FURTHER

**1** page 67
**Answer: (a) 15 (b) 6 (c) 8**
**Why:** (a) 3 + 12 = 15, so 15:45. (b) 18 - 12 = 6, so 6:10 pm. (c) 20 - 12 = 8, so 8:05 pm.

**2** page 67
**Answer: (a) 200 (b) 400**
**Why:** Scale 1: take 2 cups off both sides. 1 box balances 2 cups. Scale 2: swap the box for 2 cups: 3 cups balance 600 g, so a cup is 200 g. The box is 2 cups: 400 g. Check scale 1: 400 + 200 + 200 = 800 = 4 × 200.

**3** page 67
**Answer: Open. Example: Scale 1: 1 apple balances 4 lemons. Scale 2: 1 lemon balances a 50 g weight. Then 1 apple is 4 × 50 = 200 g.**
**Why:** Example: a lemon is 50 g and an apple is 4 lemons, so the apple is 200 g. Another way: 1 apple balances a 200 g weight, with a second scale for a lemon.

**4** page 67
**Answer: (a) 11, 12 (b) 12, 00**
**Why:** (a) 7 am to 11 am is 4 hours: 4 × 3 = 12 minutes fast, 11:12 am. (b) At the right time 12:00 pm the clock is 5 × 3 = 15 minutes fast, so it shows 12:15 pm.

### CHECK YOURSELF

**1** page 68
**Answer: 7:25 pm**
**Why:** 6:50 to 7:00 is 10 minutes. 25 more minutes: 7:25 pm.

**2** page 68
**Answer: 2300**
**Why:** 2 kg is 2000 g. 2000 + 300 = 2300 g.

**3** page 68
**Answer: 600**
**Why:** 3 pears: 3 × 200 = 600 g. The melon is 600 g.

**4** page 68
**Answer: 160**
**Why:** 3 m is 300 cm. 300 - 140 = 160 cm.

## Unit 10 · Puzzle day

**Your turn** page 69
**Answer: 30**
**Why:** Each picture is 5 children. 6 pictures: 5, 10, 15, 20, 25, 30. So 30 children chose maths.

## START HERE

**1** page 70

**Answer: (a) 15 (b) 10 (c) 90**
**Why:** Each picture stands for 5 children. (a) Chess has 3 pictures: 3 × 5 = 15. (b) Football has 2 more pictures than swimming: 2 × 5 = 10 more. (c) 6 + 4 + 3 + 5 = 18 pictures, and 18 × 5 = 90 children.

**2** page 70

**Answer: (a) 3 (b) 15**
**Why:** 75 children is 75 ÷ 5 = 15 pictures altogether. The other rows have 5 + 3 + 4 = 12 pictures. So Art needs 15 - 12 = 3 pictures, which is 15 children.

**3** page 70

**Answer: 55**
**Why:** Numbers between 40 and 60 in the 5 times table: 45, 50, 55. Digit sums: 4 + 5 = 9, 5 + 0 = 5, 5 + 5 = 10. So the number is 55.

**4** page 71

**Answer: (a) 12 (b) 5 (c) 23**
**Why:** Count in 5s for each group with a slash, then add the strokes. Dogs: 5 + 5 + 2 = 12. Cats: 5 + 2 = 7. Fish: 4. (b) 12 - 7 = 5. (c) 7 + 12 + 4 = 23.

**5** page 71

**Answer: (a) 507, 570, 705, 750 (b) 750 and 507 (c) 2**
**Why:** 0 cannot be first. Start with 5: 507, 570. Start with 7: 705, 750. (b) Greatest 750, smallest 507. (c) 705 and 750 are between 600 and 800: 2 numbers.

**6** page 71

**Answer: (a) 80¢, 90¢, 100¢, 110¢ (b) 1 coin of 10¢ and 5 coins of 20¢**
**Why:** Each swap of a 10¢ for a 20¢ adds 10¢: 60, 70, 80, 90, 100, 110. $1.10 is the last column: 1 coin of 10¢ and 5 coins of 20¢. Check: 10 + 100 = 110 and 1 + 5 = 6 coins.

**7** page 72

**Answer: (a) banana (b) mango (c) apple**
**Why:** Mei has the mango, so cross out mango for Ali and Ravi. Ali does not have the apple and not the mango, so Ali has the banana. Ravi has the apple.

**8** page 72

**Answer: (a) 3/6 (b) 3/6 (c) 300**
**Why:** (a) 1/6 + 2/6 = 3/6. (b) 6/6 - 3/6 = 3/6. (c) 3 slices left, 3 × 100 g = 300 g.

**9** page 72

**Answer: (a) 3.60 (b) 1.40**
**Why:** (a) $1.20 + $1.20 + $1.20 = $3.60. (b) $5 - $3.60 = $1.40. Count on: $3.60 to $4 is 40¢, then $1 more makes $5.

**10** page 72

**Answer: (a) 40 (b) 12**
**Why:** (a) 4 rows of 10: 4 × 10 = 40. (b) 40 - 28 = 12 empty chairs.

## THINK HARDER

**1** page 73

**Answer: (a) tennis (b) chess (c) art (d) swimming**
**Why:** Siti is not swimming or tennis, and chess is taken, so Siti has art. Ali is not chess or art, so Ali has swimming or tennis. Ben is not tennis, so Ben has swimming. Ali has tennis.

**2** page 73
**Answer: (a) 5 (b) 35 (c) 15**
**Why:** Apples have 7 - 3 = 4 more pictures than pears. 4 pictures are 20 children, so 1 picture is 20 ÷ 4 = 5 children. Apples: 7 × 5 = 35. Pears: 3 × 5 = 15. Check: 35 - 15 = 20.

**3** page 73
**Answer: (a) 5 (b) 5**
**Why:** Pump A adds 2, 4, 6, 8, 10 so its totals are 2, 6, 12, 20, 30. Pump B adds 5 a minute: 5, 10, 15, 20, 25. At minute 4 they are level at 20 l. At minute 5 pump A has 30 l and pump B has 25 l, so pump A has 5 l more.

## GO FURTHER

**1** page 74
**Answer: (a) 35 (b) 25 (c) 10**
**Why:** (a) 3 pictures are 30 and half a picture is 5: 35. (b) 2 pictures are 20 and half a picture is 5: 25. (c) 35 - 25 = 10.

**2** page 74
**Answer: (a) 11 (b) No**
**Why:** All 4 coins are worth 5 + 10 + 20 + 20 = 55¢, so 60¢ is too much. The amounts are 5, 10, 15, 20, 25, 30, 35, 40, 45, 50 and 55: 11 amounts. Every multiple of 5 from 5¢ to 55¢ works.

**3** page 74
**Answer: Open. Example: Ali, Mei and Ravi have a cat, a dog and a fish. Mei has the fish. Ali does not have the cat. Then Ali has the dog and Ravi has the cat.**
**Why:** A good puzzle has exactly one answer. Example: Mei has the fish (so Ali and Ravi have the cat and the dog). Ali does not have the cat, so Ali has the dog and Ravi has the cat. Try to find a second answer: there is none.

**4** page 74
**Answer: (a) 3 small and 5 large (b) No**
**Why:** 8 small stickers cost $2.40. Each swap to a large adds 20¢: $2.60, $2.80, $3.00, $3.20, $3.40. That is 5 swaps: 5 large, 3 small. Totals rise in 20¢ steps, so $3.30 never happens.

## CHECK YOURSELF

**1** page 75
**Answer: 50**
**Why:** Tuesday has 5 pictures. Each picture is 10 children. 5 × 10 = 50 children.

**2** page 75
**Answer: 4**
**Why:** 0 cannot be first. Start with 1: 104, 140. Start with 4: 401, 410. That is 4 numbers.

**3** page 75
**Answer: red**
**Why:** Mei has blue. Ali does not have red, so Ali has green. Ben has red.

**4** page 75
**Answer: 7/12**
**Why:** 12/12 - 5/12 = 7/12 is left.